

taking $I_E \approx I_C$ we have

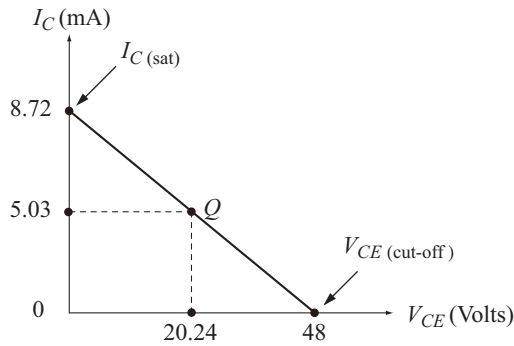
$$I_{C(\text{sat})} = \frac{48 \text{ V}}{3.3 \text{ k}\Omega + 2.2 \text{ k}\Omega} = 8.72 \text{ mA}$$

(e) DC load line

When transistor is at cut-off, $I_C \approx 0$

From Equation (B) we have

$$V_{CE(\text{cut-off})} = 48 \text{ V}$$



Example 2.26

Determine R_C and R_B for a fixed bias configuration using the following data

$$\begin{aligned} V_{CC} &= 12 \text{ V} & I_{CQ} &= 2.5 \text{ mA} \\ V_{CEQ} &= 6 \text{ V} & \beta &= 80 \end{aligned}$$

Solution

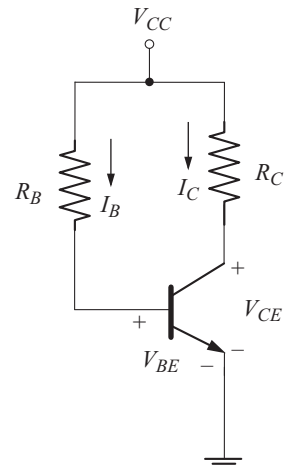
Calculation of R_C

From the KVL equation of collector emitter circuit we have

$$\begin{aligned} V_{CC} &= I_C R_C + V_{CE} \\ R_C &= \frac{V_{CC} - V_{CE}}{I_C} = \frac{12 \text{ V} - 6 \text{ V}}{2.5 \text{ mA}} = 2.4 \text{ k}\Omega \end{aligned}$$

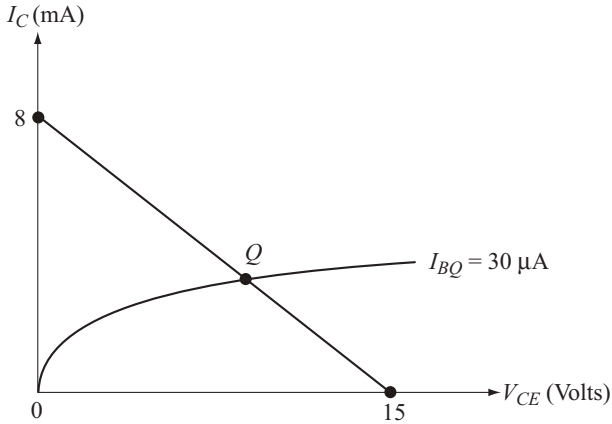
Calculation of R_B

$$\begin{aligned} R_B &= \frac{V_{CC} - V_{BE}}{I_B} \\ I_B &= \frac{I_C}{\beta} = \frac{2.5 \text{ mA}}{80} = 31.25 \mu\text{A} \\ R_B &= \frac{12 \text{ V} - 0.7 \text{ V}}{31.25 \mu\text{A}} = 361.6 \text{ k}\Omega \end{aligned}$$

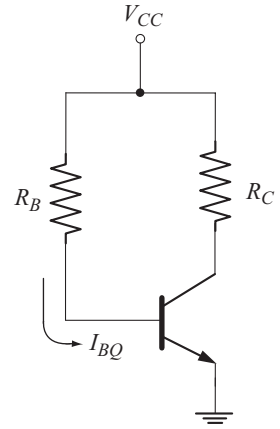


Example 2.27

Using the transistor output characteristics of Fig. (a), find the values of V_{CC} , R_B and R_C for the fixed bias configuration of Fig. (b).



(a)



(b)

Solution

From the characteristics of Fig. (a) we get the following data

$$V_{CC} = 15 \text{ V} \quad I_{BQ} = I_B = 30 \text{ } \mu\text{A} \quad I_{C(\text{sat})} = 8 \text{ mA}$$

$$I_{C(\text{sat})} = \frac{V_{CC}}{R_C}$$

$$\therefore$$

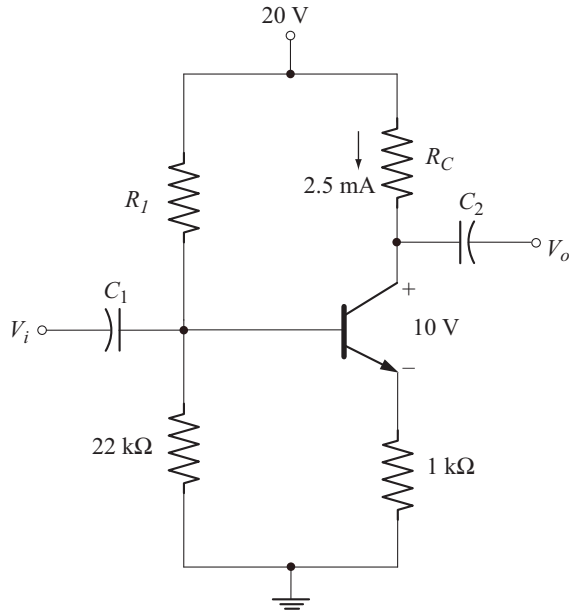
$$\begin{aligned} R_C &= \frac{V_{CC}}{I_{C(\text{sat})}} \\ &= \frac{15 \text{ V}}{8 \text{ mA}} = 1.875 \text{ k}\Omega \end{aligned}$$

From the KVL equation of base-emitter circuit, we have

$$\begin{aligned} V_{CC} &= I_B R_B + V_{BE} \\ R_B &= \frac{V_{CC} - V_{BE}}{I_B} \\ &= \frac{20 \text{ V} - 0.7 \text{ V}}{30 \text{ } \mu\text{A}} \\ &= 643.33 \text{ k}\Omega \end{aligned}$$

Example 2.28

For the circuit shown below determine the values of R_1 and R_C .

**Solution**

Given

$$I_{CQ} = I_C = 2.5 \text{ mA} \quad V_{CEQ} = V_{CE} = 10 \text{ V}$$

Calculation of R_1

$$\begin{aligned} V_B &= V_{BE} + V_E \\ V_E &= I_E R_E \approx I_C R_E = (2.5 \text{ mA})(1 \text{ k}\Omega) = 2.5 \text{ V} \\ \therefore V_B &= 0.7 \text{ V} + 2.5 \text{ V} = 3.2 \text{ V} \end{aligned}$$

$$\text{But } V_B = V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2}$$

$$3.2 \text{ V} = \frac{(20 \text{ V})(22 \text{ k}\Omega)}{R_1 + 22 \text{ k}\Omega}$$

$$R_1 + 22 \text{ k}\Omega = 137.5 \text{ k}\Omega$$

$$R_1 = 115.5 \text{ k}\Omega$$

Calculation of R_C

$$I_C = \frac{V_{CC} - V_C}{R_C}$$

$$\Rightarrow R_C = \frac{V_{CC} - V_C}{I_C} \quad (\text{A})$$

$$V_C = V_{CE} + V_E = 10 \text{ V} + 2.5 \text{ V} = 12.5 \text{ V}$$

Now from Equation (A)

$$R_C = \frac{20 \text{ V} - 12.5 \text{ V}}{2.5 \text{ mA}} = 3 \text{ k}\Omega$$

Example 2.29

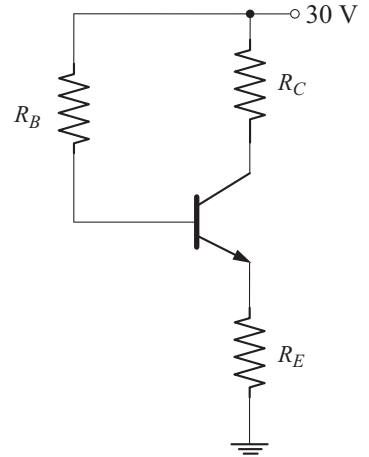
For the emitter bias circuit shown below find the values of R_C , R_E and R_B using the following specifications.

$$I_{C(\text{sat})} = 10 \text{ mA}$$

$$I_{CQ} = \frac{1}{2} I_{C(\text{sat})}$$

$$V_C = 20 \text{ V}$$

Assume silicon transistor with $\beta = 100$.



Solution

Calculation of R_C

$$I_{CQ} = I_C = \frac{1}{2} I_{C(\text{sat})} = 5 \text{ mA}$$

$$R_C = \frac{V_{CC} - V_C}{I_C} = \frac{30 \text{ V} - 20 \text{ V}}{5 \text{ mA}} = 2 \text{ k}\Omega$$

Calculation of R_E

$$I_{C(\text{sat})} = \frac{V_{CC}}{R_C + R_E}$$

$$R_C + R_E = \frac{V_{CC}}{I_{C(\text{sat})}} = \frac{30 \text{ V}}{10 \text{ mA}} = 3 \text{ k}\Omega$$

$$R_E = 3 \text{ k}\Omega - R_C = 1 \text{ k}\Omega$$

Calculation of R_B

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta)R_E} \quad (\text{A})$$

$$I_B = \frac{I_C}{\beta} = \frac{5 \text{ mA}}{100} = 50 \mu\text{A}$$

Using this value in Equation (A) we have,

$$50 \mu\text{A} = \frac{30 \text{ V} - 0.7 \text{ V}}{R_B + (101)(1 \text{ k}\Omega)}$$

$$R_B + 101 \text{ k}\Omega = 586 \text{ k}\Omega$$

$$R_B = 586 \text{ k}\Omega - 101 \text{ k}\Omega = 485 \text{ k}\Omega$$

Example 2.30

Design an emitter stabilized network shown using the following data

$$I_{CQ} = \frac{1}{2} I_{C(\text{sat})}$$

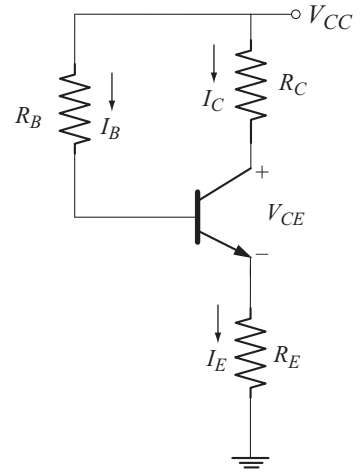
$$V_{CEQ} = \frac{1}{2} V_{CC}$$

$$V_{CC} = 20 \text{ V}$$

$$I_{C(\text{sat})} = 10 \text{ mA}$$

$$\beta = 120$$

$$R_C = 4 R_E$$



Solution

$$I_C = I_{CQ} = \frac{1}{2} I_{C(\text{sat})} = \frac{1}{2} (10 \text{ mA}) = 5 \text{ mA}$$

$$I_B = \frac{I_C}{\beta} = \frac{5 \text{ mA}}{120} = 41.66 \mu\text{A}$$

$$V_{CEQ} = V_{CE} = \frac{1}{2} V_{CC} = \frac{1}{2} (20 \text{ V}) = 10 \text{ V}$$

Calculation of R_C and R_E

From the KVL equation of collector-emitter circuit we have

$$V_{CC} = I_C R_C + V_{CE} + I_E R_E$$

Given $R_C = 4 R_E$ and taking $I_C \approx I_E$ we have

$$V_{CC} - V_{CE} = I_C [5R_E]$$

$$R_E = \frac{V_{CC} - V_{CE}}{5I_C} = \frac{20 \text{ V} - 10 \text{ V}}{5(5 \text{ mA})} = 400 \Omega$$

$$R_C = 4 R_E = 1.6 \text{ k}\Omega$$

Calculation of R_B

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta)R_E}$$

$$41.66 \mu\text{A} = \frac{20 \text{ V} - 0.7 \text{ V}}{R_B + 121(0.4 \text{ k}\Omega)}$$

$$R_B + 48.4 \text{ k}\Omega = \frac{19.3 \text{ V}}{41.66 \mu\text{A}} = 463.27 \text{ k}\Omega$$

$$R_B = 463.27 \text{ k}\Omega - 48.4 \text{ k}\Omega = 414.87 \text{ k}\Omega$$

◆ 2.10 DESIGN RULE

To design the values of biasing circuit elements R_1 , R_2 and R_E , the following thumb rules are normally used.

- In emitter stabilized and voltage divider configurations the emitter to ground voltage V_E is taken equal to one-tenth of the dc supply voltage V_{CC}

$$\text{i.e.,} \quad V_E = \frac{1}{10} V_{CC} \quad (2.54)$$

R_E is calculated using the relation

$$R_E = \frac{V_E}{I_E} \approx \frac{V_E}{I_C} \quad (2.55)$$

This rule ensures that, R_E will neither be too large nor be too small. Too large value of R_E reduces the swing of V_{CE} (i.e., output voltage swing) and too small value of R_E degrades the stability of Q point.

- In voltage divider configuration, the resistors R_1 and R_2 are calculated assuming that the current through R_1 and R_2 is at least 10 times greater than the base current. This leads to the condition

$$R_2 \leq \frac{1}{10} \beta R_E \quad (2.56)$$

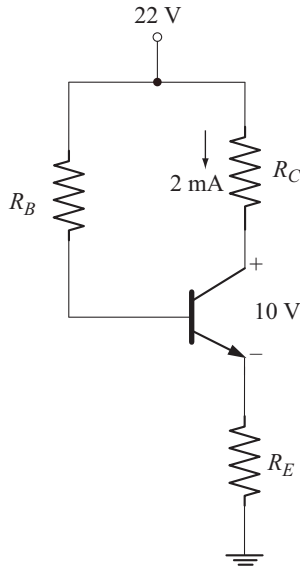
R_1 is calculated from the relation

$$V_B = V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} \quad (2.57)$$

Design of biasing networks is considered in the following examples.

Example 2.31

Design the values of R_B , R_E and R_C for the emitter-stabilized bias circuit shown below. Assume silicon transistor with $\beta = 140$.

**Solution**

Given

$$I_{CQ} = I_C = 2 \text{ mA}$$

$$V_{CEQ} = V_{CE} = 10 \text{ V} \quad \beta = 140$$

There are three unknowns, R_B , R_E and R_C . But we have only the following two equations.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta)R_E} \quad (\text{A})$$

$$V_{CC} = I_C R_C + V_{CE} + \underbrace{I_E R_E}_{V_E} \quad (\text{B})$$

To start with the design, we assume V_E . Typically V_E is taken equal to one-tenth of V_{CC} .

$$V_E = \frac{V_{CC}}{10} = \frac{22 \text{ V}}{10} = 2.2 \text{ V}$$

$$V_E = I_E R_E \approx I_C R_E$$

$$\therefore R_E = \frac{V_E}{I_C} = \frac{2.2 \text{ V}}{2 \text{ mA}} = 1.1 \text{ k}\Omega$$

$$I_B = \frac{I_C}{\beta} = \frac{2 \text{ mA}}{140} = 14.28 \mu\text{A}$$

From Equation (A)

$$14.26 \mu\text{A} = \frac{22 \text{ V} - 0.7 \text{ V}}{R_B + (141)(1.1 \text{ k}\Omega)}$$

$$R_B + 155.1 \text{ k}\Omega = 1493.68 \text{ k}\Omega$$

$$R_B = 1.33 \text{ M}\Omega$$

From Equation (B)

$$R_C = \frac{V_{CC} - V_{CE} - V_E}{I_C} = \frac{22 \text{ V} - 10 \text{ V} - 2.2 \text{ V}}{2 \text{ mA}} = 4.9 \text{ k}\Omega$$

Example 2.32

Determine the values of R_C , R_E , R_1 and R_2 for the current-gain stabilized (Beta-independent) circuit with $I_C = 10 \text{ mA}$, $V_{CE} = 12 \text{ V}$ and $V_{CC} = 24 \text{ V}$. Assume silicon transistor with $\beta_{(\min)} = 100$.

Solution

Current-gain stabilized (Beta-independent) circuit is nothing but voltage divider bias circuit.
Given

$$I_{CQ} = I_C = 10 \text{ mA} \quad V_{CEQ} = V_{CE} = 12 \text{ V} \quad \beta_{(\min)} = \beta = 100$$

Calculation of R_E

Let

$$V_E = \frac{V_{CC}}{10}$$

$$\therefore V_E = \frac{24 \text{ V}}{10} = 2.4 \text{ V}$$

$$R_E = \frac{V_E}{I_E} \approx \frac{V_E}{I_C} = \frac{2.4 \text{ V}}{10 \text{ mA}} = 240 \Omega$$

Calculation of R_C

$$I_C = \frac{V_{CC} - V_C}{R_C}$$

$$\Rightarrow R_C = \frac{V_{CC} - V_C}{I_C}$$

$$V_C = V_{CE} + V_E = 12 \text{ V} + 2.4 \text{ V} = 14.4 \text{ V}$$

$$R_C = \frac{24 \text{ V} - 14.4 \text{ V}}{10 \text{ mA}} = 960 \Omega$$

Calculation R_1 and R_2

$$V_B = \frac{V_{CC} R_2}{R_1 + R_2} \tag{A}$$

$$V_B = V_{BE} + V_E = 0.7 \text{ V} + 2.4 \text{ V} = 3.1 \text{ V}$$

In Equation (A), we have two unknowns R_1 and R_2 .

Let us assume that the current through R_1 and R_2 is 10 times more than I_B . This implies that

$$R_2 \leq \frac{\beta R_E}{10}$$

$$R_2 \leq \frac{(100)(240 \Omega)}{10}$$

$$R_2 \leq 2.4 \text{ k}\Omega$$

$$R_2 = 2.4 \text{ k}\Omega$$

Let

From Equation (A)

$$3.1 \text{ V} = \frac{(24 \text{ V})(2.4 \text{ k}\Omega)}{R_1 + 2.4 \text{ k}\Omega}$$

$$R_1 + 2.4 \text{ k}\Omega = 18.58 \text{ k}\Omega$$

$$R_1 = 16.18 \text{ k}\Omega$$

◆ 2.11 CIRCUITS WITH PNP TRANSISTORS

So far we have analysed biasing configurations using *nnp* transistors. Now let us consider the circuits with *pnnp* transistor.

2.11.1 Emitter-stabilized Configuration using pnp Transistor

Figure 2.21 shows emitter-stabilized configuration with *pnnp* transistor.

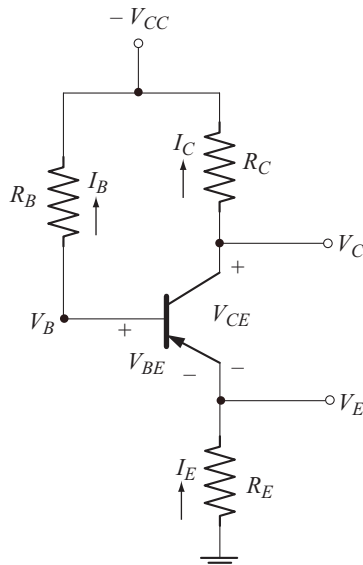


Fig. 2.21 Emitter-stabilized configuration using pnp transistor

It is necessary to make the following observations regarding the current directions and voltage polarities employed for *pnp* transistor in contrast with those of *nnp* transistor.

- The directions of I_B , I_C and I_E in *pnp* transistor are exactly opposite to that of *nnp* transistor. Since the correct directions of currents are taken in Fig. 2.21, numerical values of these currents will be positive.
- The polarities of V_{CE} , V_{BE} , V_C , V_E etc are retained as in the case of *nnp* transistor. Since the actual polarities are the other way, numerical values of these voltages will be negative.
- *pnp* transistor requires negative dc supply $-V_{CC}$ in the collector circuit.

Circuit Analysis

Applying KVL to base-emitter circuit, we have

$$-I_E R_E + V_{BE} - I_B R_B + V_{CC} = 0$$

Using $I_E = (1 + \beta) I_B$, we obtain

$$I_B = \frac{V_{CC} + V_{BE}}{R_B + (1 + \beta) R_E} \quad (2.58)$$

It should be note that, V_{CC} is positive and V_{BE} is -0.7 V for *pnp* silicon transistor. This results in positive value for I_B .

Applying KVL to collector, emitter circuit we have

$$\begin{aligned} -I_E R_E + V_{CE} - I_C R_C + V_{CC} &= 0 \\ V_{CE} &= -V_{CC} + I_C R_C + I_E R_E \end{aligned} \quad (2.59)$$

Since the direction of I_E is reversed but the polarity of V_E is retained

$$V_E = -I_E R_E \quad (2.60)$$

As in case of *nnp* transistor

$$V_B = V_{BE} + V_E \quad (2.61)$$

$$V_C = V_{CE} + V_E \quad (2.62)$$

and

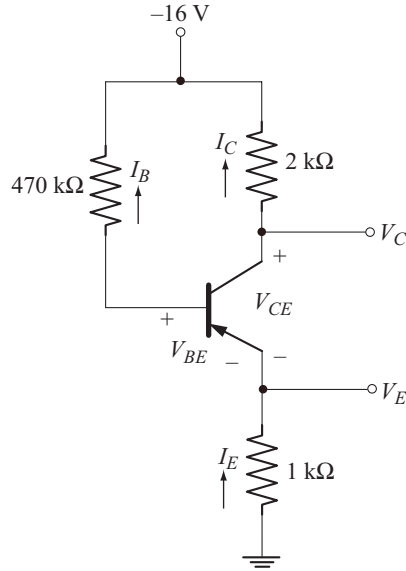
$$V_{CB} = V_C - V_B \quad (2.63)$$

Example 2.33

For the emitter-stabilized bias configuration shown below determine

- I_B , I_C and I_E
- V_E and V_B
- V_{CE} and V_C
- V_{CB} and $I_{C(sat)}$

Assume silicon transistor with $\beta = 75$.

**Solution**

The circuit uses *pn*p silicon transistor

$$\therefore V_{BE} = -0.7 \text{ V}$$

also

$$-V_{CC} = -16 \text{ V} \Rightarrow V_{CC} = 16 \text{ V}$$

(a) I_B , I_C and I_E

$$\begin{aligned} I_B &= \frac{V_{CC} + V_{BE}}{R_B + (1 + \beta)R_E} \\ &= \frac{16 \text{ V} - 0.7 \text{ V}}{470 \text{ k}\Omega + (76)(1 \text{ k}\Omega)} = 28.02 \text{ }\mu\text{A} \end{aligned}$$

$$I_C = \beta I_B = (75)(28.02 \text{ }\mu\text{A}) = 2.1 \text{ mA}$$

$$I_E = I_C + I_B = 2.1 \text{ mA} + 28.02 \text{ }\mu\text{A} = 2.12 \text{ mA}$$

(b) V_E and V_B

$$V_E = -I_E R_E = -(2.12 \text{ mA})(1 \text{ k}\Omega) = -2.12 \text{ V}$$

$$V_B = V_{BE} + V_E = -0.7 \text{ V} - 2.12 \text{ V} = -2.82 \text{ V}$$

(c) V_{CE} and V_C

$$\begin{aligned} V_{CE} &= -V_{CC} + I_C R_C + I_E R_E \\ &= -16 \text{ V} + (2.1 \text{ mA})(2 \text{ k}\Omega) + (2.12 \text{ mA})(1 \text{ k}\Omega) \\ &= -9.68 \text{ V} \end{aligned}$$

$$V_C = V_{CE} + V_E = -9.68 \text{ V} - 2.12 \text{ V} = -11.8 \text{ V}$$

(d) V_{CB} and $I_{C(sat)}$

$$V_{CB} = V_C - V_B = -11.8 \text{ V} - (-2.82 \text{ V}) = -8.98 \text{ V}$$

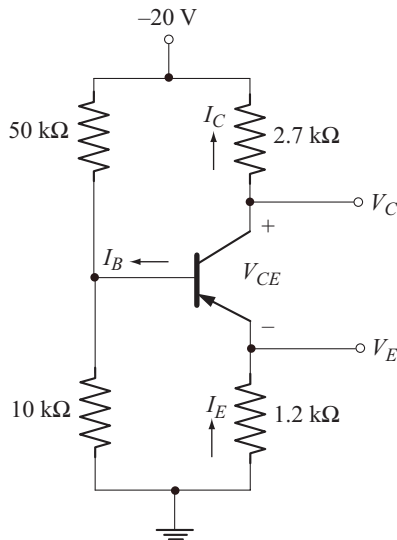
$$I_{C(sat)} = \frac{V_{CC}}{R_C + R_E} = \frac{16 \text{ V}}{2 \text{ k}\Omega + 1 \text{ k}\Omega} = 5.33 \text{ mA}$$

Example 2.34

For the voltage divider configuration shown below calculate

- (a) I_B , I_C , and I_E
 (b) V_E and V_B
 (c) V_{CE} and V_C
 (d) V_{CB} and $I_{C(sat)}$

Assume silicon transistor with $\beta = 110$.

**Solution**

$$-V_{CC} = -20 \text{ V} \Rightarrow V_{CC} = 20 \text{ V}$$

$$V_{BE} = -0.7 \text{ V} \quad (\text{pnp Si transistor})$$

(a) I_B , I_C , and I_E

$$I_B = \frac{V_{Th} + V_{BE}}{R_{Th} + (1 + \beta)R_E} \quad (\text{A})$$

$$V_{Th} = \frac{V_{CC} R_2}{R_1 + R_2} = \frac{(20 \text{ V})(10 \text{ k}\Omega)}{50 \text{ k}\Omega + 10 \text{ k}\Omega} = 3.33 \text{ V}$$

$$R_{Th} = R_1 \parallel R_2 = 50 \text{ k}\Omega \parallel 10 \text{ k}\Omega = 8.33 \text{ k}\Omega$$

$$I_B = \frac{3.33 \text{ V} - 0.7 \text{ V}}{8.33 \text{ k}\Omega + (111)(1.2 \text{ k}\Omega)} = 18.58 \mu\text{A}$$

$$I_C = \beta I_B = (110)(18.58 \mu\text{A}) = 2.04 \text{ mA}$$

$$I_E = I_C + I_B = 2.05 \text{ mA}$$

(b) V_E and V_B

$$V_E = -I_E R_E = -(2.05 \text{ mA})(1.2 \text{ k}\Omega) = -2.46 \text{ V}$$

$$V_B = V_{BE} + V_E = -0.7 \text{ V} - 2.46 \text{ V} = -3.16 \text{ V}$$

(c) V_{CE} and V_C

$$\begin{aligned} V_{CE} &= -V_{CC} + I_C R_C + I_E R_E \\ &= -20 \text{ V} + (2.04 \text{ mA})(2.7 \text{ k}\Omega) + (2.05 \text{ mA})(1.2 \text{ k}\Omega) \\ &= -12.03 \text{ V} \end{aligned}$$

$$V_C = V_{CE} + V_E = -12.03 \text{ V} - 2.46 \text{ V} = -14.49 \text{ V}$$

(d) V_{CB} and $I_{C(\text{sat})}$

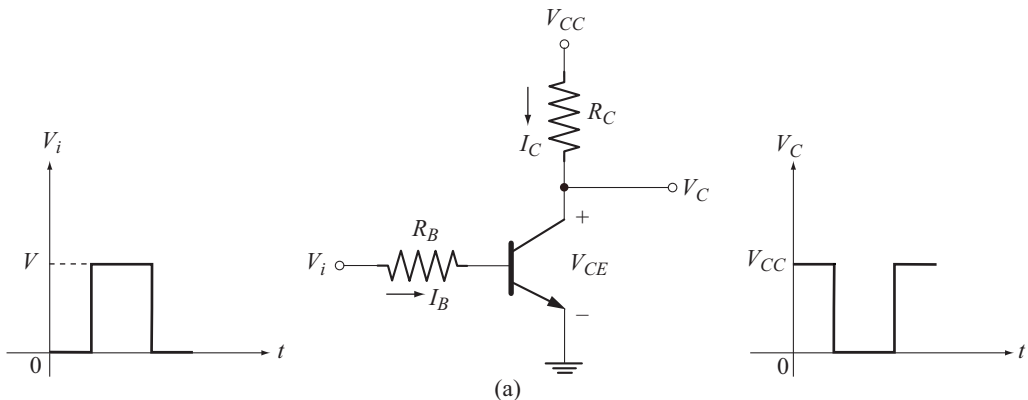
$$V_{CB} = V_C - V_B = -14.49 \text{ V} - (-3.16 \text{ V}) = -11.33 \text{ V}$$

$$I_{C(\text{sat})} = \frac{V_{CC}}{R_C + R_E} = \frac{20 \text{ V}}{2.7 \text{ k}\Omega + 1.2 \text{ k}\Omega} = 5.13 \text{ mA}$$

◆ 2.12 TRANSISTOR SWITCHING NETWORKS

When biased in the middle of active region, transistor works as a faithful amplifier. So far we have studied biasing configurations which bias the transistor in the active region. When operated between cut-off and saturation, transistor works as a switch or an inverter. Such a switching circuit is useful in computers and control applications

Fig. 2.22(a) shows the transistor switch and the output characteristics of the transistor is shown in Fig. 2.22(b).



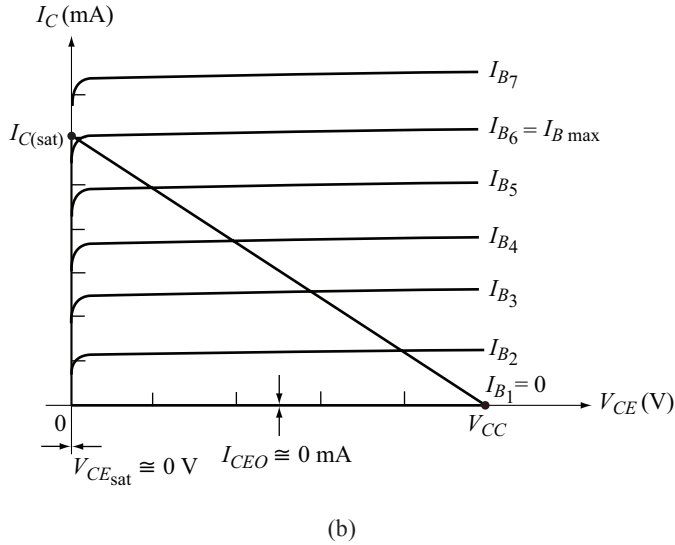


Fig. 2.22 (a) Transistor switch (b) Output characteristics

Applying KVL to the base-emitter circuit, we have

$$V_i = I_B R_B + V_{BE}$$

$$I_B = \frac{V_i - V_{BE}}{R_B} \quad (2.64)$$

For the collector, emitter circuit

$$V_{CC} = I_C R_C + V_{CE}$$

$$I_C = \frac{V_{CC} - V_{CE}}{R_C} \quad (2.65)$$

The output voltage is

$$V_C = V_{CE} \quad (2.66)$$

Now let us consider the following cases.

Case (i) : When $V_i = V$

This high input voltage will turn on the transistor. Let us assume that, R_B is properly selected so as to produce heavy base current which drives the transistor in to saturation.

$$\text{In saturation,} \quad V_{CE} = V_{CE(\text{sat})}$$

$$\text{and} \quad I_C = I_{C(\text{sat})}$$

For silicon transistor $V_{CE(\text{sat})}$ is typically 0.1 V – 0.3 V.

From Equation (2.65), the collector current at saturation is given by

$$I_{C(\text{sat})} = \frac{V_{CC} - V_{CE(\text{sat})}}{R_C} \quad (2.67)$$

and from Equation (2.66), the output voltage is

$$V_C = V_{CE(\text{sat})} \quad (2.68)$$

From Equation (2.64), the base current is

$$I_B = \frac{V - V_{BE}}{R_B} \quad (2.69)$$

Just before saturation, the transistor is in the active region. The base current in the active region just before saturation is approximately given by

$$I_{B(\text{max})} \approx \frac{I_{C(\text{sat})}}{\beta_{dc}} \quad (2.70)$$

Some times β_{dc} is also denoted by h_{FE} .

To ensure saturation, the level of I_B given by Equation (2.69) must be greater than $I_{B(\text{max})}$ given in Equation (2.70).

$$\text{i.e.,} \quad I_B > \frac{I_{C(\text{sat})}}{\beta_{dc}} \quad (2.71)$$

Typically I_B is taken equal to 120–150% of $I_{B(\text{max})}$ to ensure saturation. The resistance between collector and emitter terminals, at saturation is

$$R_{(\text{sat})} = \frac{V_{CE(\text{sat})}}{I_{C(\text{sat})}} \quad (2.72)$$

Under ideal conditions, $V_{CE(\text{sat})} \approx 0 \text{ V}$

From Equation (2.67),

$$I_{C(\text{sat})} \approx \frac{V_{CC}}{R_C} \quad (2.73)$$

From Equation (2.66),

$$V_C \approx 0 \text{ V} \quad (2.74)$$

and from Equation (2.72)

$$R_{(\text{sat})} \approx 0 \Omega \quad (2.75)$$

Note that, when transistor operates in saturation, there exists a short circuit between collector and emitter terminals.

Figure 2.23 shows the representation of saturated transistor.

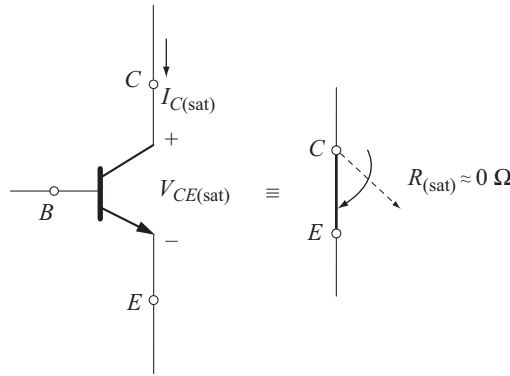


Fig. 2.23 Representation of saturated transistor

Case (ii) : When $V_i = 0$

When $V_i = 0$, $I_B = 0$, only leakage current I_{CEO} flows from collector to emitter and the transistor is driven into cut-off.

$$\therefore I_C = I_{CEO} \quad (2.76)$$

I_{CEO} is in the order of few microamperes.

From Equation (2.65), the output voltage is

$$V_C = V_{CE(\text{cut-off})} = V_{CC} - I_{CEO} R_C \quad (2.77)$$

The resistance between collector and emitter terminals, at cut-off is

$$R_{(\text{cut-off})} = \frac{V_{CE(\text{cut-off})}}{I_{CEO}} \quad (2.78)$$

Under ideal conditions, $I_{CEO} \approx 0$.

Now from Equation (2.77),

$$V_C = V_{CE(\text{cut-off})} = V_{CC} \quad (2.79)$$

and from Equation (2.78),

$$R_{(\text{cut-off})} = \frac{V_{CC}}{0} = \infty \Omega \quad (2.80)$$

Observe that, there exists an open circuit between collector and emitter terminals when the transistor operates in the cut-off region.

The representation of transistor when operates in the cut-off region is shown in Fig. 2.24.

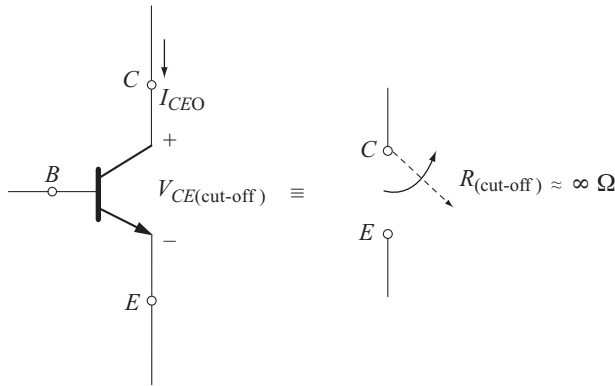


Fig. 2.24 Representation of transistor when operating at cut-off

Summary

A high input voltage drives the transistor into saturation. When in saturation, the transistor

- Conducts heavy collector current, $I_{C(sat)} \approx \frac{V_{CC}}{R_C}$
- Has low voltage between collector and emitter terminals

$$V_C = V_{CE(sat)} \approx 0 \text{ V}$$

- Has zero resistance (short circuit) from collector to emitter
- Acts as a closed switch

A low input voltage (0 V) drives the transistor into cut-off. When at cut-off the transistor

- Does not conduct, $I_C = I_{CEO} \approx 0$
- has high voltage between collector and emitter terminals

$$V_E = V_{CE} \approx V_{CC}$$

- has infinite resistance (open circuit) from collector to emitter
- acts as an opened switch

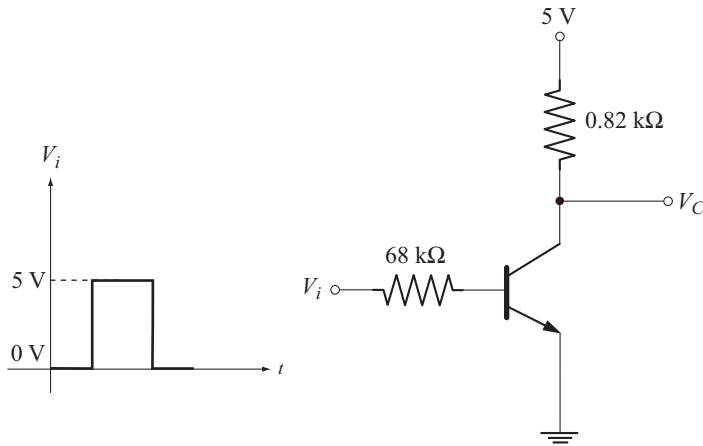
Observe that a high input voltage results in low output voltage and vice-versa. Hence the transistor switch is nothing but an inverter.

Example 2.35

Figure given below shows the transistor switch

- Check whether the circuit works properly
- Determine the output voltage levels at cut-off and saturation
- Calculate $R_{(sat)}$ and $R_{(cut-off)}$

$$\begin{aligned} \text{Take } V_{BE} &= 0.7 \text{ V} & V_{CE(sat)} &= 0.15 \text{ V} \\ I_{CEO} &= 10 \mu\text{A} & \beta_{dc} = h_{FE} &= 125 \end{aligned}$$

**Solution**

$$(a) \quad I_{C(sat)} = \frac{V_{CC} - V_{CE(sat)}}{R_C} = \frac{5 \text{ V} - 0.15 \text{ V}}{0.82 \text{ k}\Omega} = 5.91 \text{ mA}$$

$$I_{B(max)} = \frac{I_{C(sat)}}{\beta_{dc}} = \frac{5.9 \text{ mA}}{125} = 47.2 \text{ }\mu\text{A}$$

When $V_i = 5 \text{ V}$, the actual base current is

$$I_B = \frac{V_i - V_{BE}}{R_B} = \frac{5 \text{ V} - 0.7 \text{ V}}{68 \text{ k}\Omega} = 63.23 \text{ }\mu\text{A}$$

Observe that, $I_B > I_{B(max)}$. Hence the circuit works properly.

(b) When $V_i = 5 \text{ V}$

$$V_C = V_{CE(sat)} = 0.15 \text{ V}$$

When $V_i = 0 \text{ V}$

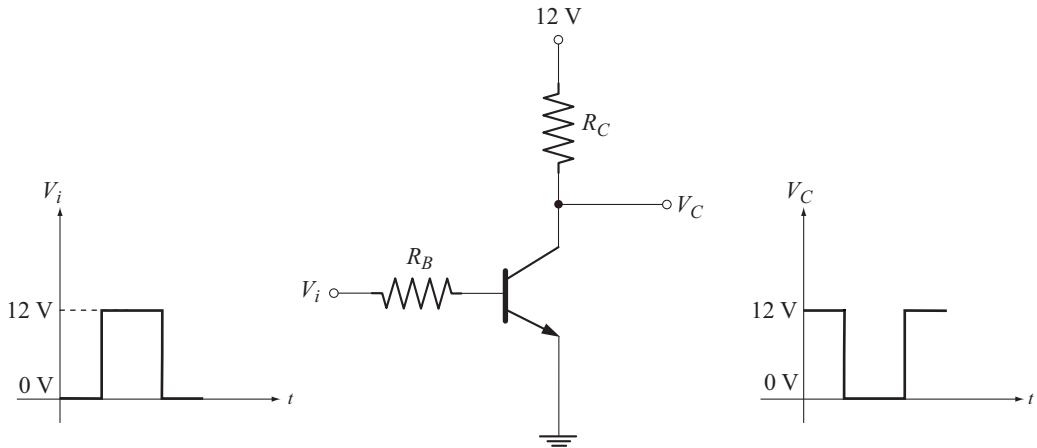
$$\begin{aligned} V_C &= V_{CE(\text{cut-off})} = V_{CC} - I_{CEO} R_C \\ &= 5 \text{ V} - (10 \text{ }\mu\text{A})(0.82 \text{ k}\Omega) \approx 5 \text{ V} \end{aligned}$$

$$(c) \quad R_{(sat)} = \frac{V_{CE(sat)}}{I_{C(sat)}} = \frac{0.15 \text{ V}}{5.9 \text{ mA}} = 25.42 \text{ }\Omega \text{ (low)}$$

$$R_{(\text{cut-off})} = \frac{V_{CE(\text{cut-off})}}{I_{CEO}} = \frac{5 \text{ V}}{10 \text{ }\mu\text{A}} = 500 \text{ k}\Omega \text{ (very high)}$$

Example 2.36

For the transistor inverter shown below, determine the values of R_B and R_C . Take $I_{C(sat)} = 12 \text{ mA}$ and $h_{FE} = 200$.



Solution

Calculation of R_C

$$I_{C(\text{sat})} = \frac{V_{CC} - V_{CE(\text{sat})}}{R_C}$$

From the output waveform, $V_{CE(\text{sat})} = 0 \text{ V}$

$$\therefore R_C = \frac{V_{CC}}{I_{C(\text{sat})}} = \frac{12 \text{ V}}{12 \text{ mA}} = 1 \text{ k}\Omega$$

Calculation of R_B

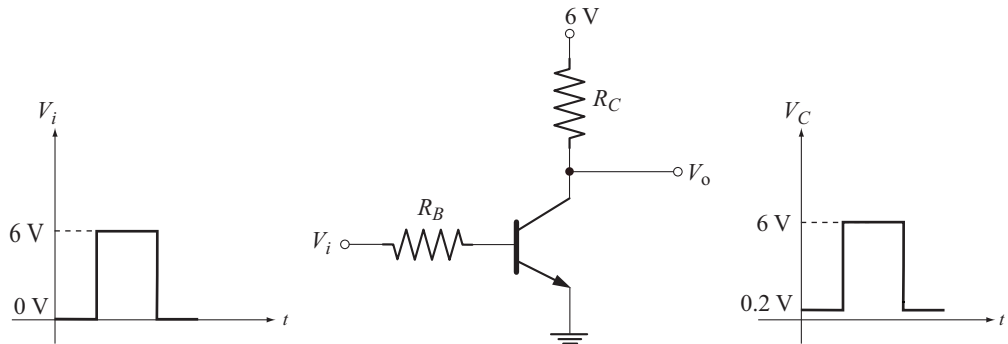
$$I_{B(\text{max})} = \frac{I_{C(\text{sat})}}{\beta_{dc}} = \frac{12 \text{ mA}}{200} = 60 \mu\text{A}$$

Let $I_B = 150 \% \text{ of } I_{B(\text{max})} \quad [\text{To ensure saturation}]$
 $= (1.5) (60 \mu\text{A}) = 90 \mu\text{A}$

$$\begin{aligned} R_B &= \frac{V_i - V_{BE}}{I_B} \\ &= \frac{12 \text{ V} - 0.7 \text{ V}}{90 \mu\text{A}} \\ &= 125.55 \text{ k}\Omega \end{aligned}$$

Example 2.37

Design the transistor inverter shown below with a saturation current of 10 mA. Take I_B equal to 120 % of $I_{B(\text{max})}$. For the transistor used $\beta_{dc} = 100$ and $V_{BE} = 0.7 \text{ V}$.

**Solution****Calculation of R_C**

$$I_{C(\text{sat})} = \frac{V_{CC} - V_{CE(\text{sat})}}{R_C}$$

From the output voltage waveform, $V_{CE(\text{sat})} = 0.2 \text{ V}$

$$R_C = \frac{V_{CC} - V_{CE(\text{sat})}}{I_{C(\text{sat})}} = \frac{6 \text{ V} - 0.2 \text{ V}}{10 \text{ mA}} = 580 \Omega$$

Calculation of R_B

$$I_{B(\text{max})} = \frac{I_{C(\text{sat})}}{\beta_{\text{dc}}} = \frac{10 \text{ mA}}{100} = 100 \mu\text{A}$$

Given

$$\begin{aligned} I_B &= 120 \% \text{ of } I_{B(\text{max})} \\ &= (1.2) (100 \mu\text{A}) = 120 \mu\text{A} \end{aligned}$$

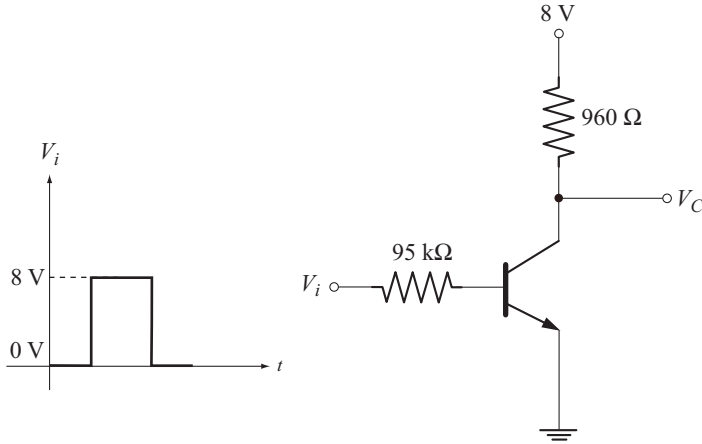
$$R_B = \frac{V_i - V_{BE}}{I_B} = \frac{6 \text{ V} - 0.7 \text{ V}}{120 \mu\text{A}} = 44.16 \text{ k}\Omega$$

Example 2.38

For the transistor switch shown below

$$V_{BE} = 0.7 \text{ V}, \quad V_{CE(\text{sat})} = 0.3 \text{ V}, \quad I_{CEO} = 5 \mu\text{A} \quad \text{and} \quad h_{FE} = 125$$

- Calculate $I_{C(\text{sat})}$ and $I_{B(\text{max})}$
- Find I_B and check whether the transistor goes to saturation when $V_i = 8 \text{ V}$
- Determine $R_{C(\text{sat})}$ and $R_{C(\text{cut-off})}$
- Sketch the output voltage waveform.

**Solution**

(a) $I_{C(\text{sat})}$ and $I_{B(\text{max})}$

$$I_{C(\text{sat})} = \frac{V_{CC} - V_{CE(\text{sat})}}{R_C} = \frac{8 \text{ V} - 0.3 \text{ V}}{960 \Omega} = 8.02 \text{ mA}$$

$$I_{B(\text{max})} = \frac{I_{C(\text{sat})}}{\beta_{\text{dc}}} = \frac{8.02 \text{ mA}}{125} = 64.16 \mu\text{A}$$

(b) I_B when $V_i = 8 \text{ V}$

$$I_B = \frac{V_i - V_{BE}}{R_B} = \frac{8 \text{ V} - 0.7 \text{ V}}{95 \text{ k}\Omega} = 76.84 \mu\text{A}$$

Since $I_B > I_{B(\text{max})}$, the transistor definitely goes to saturation when $V_i = 8 \text{ V}$.

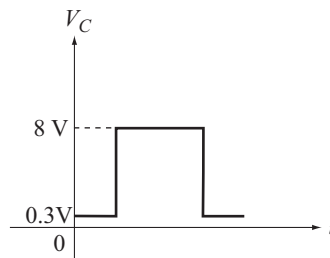
(c) $R_{(\text{sat})}$ and $R_{(\text{cut-off})}$

$$R_{(\text{sat})} = \frac{V_{CE(\text{sat})}}{I_{C(\text{sat})}} = \frac{0.3 \text{ V}}{8.02 \text{ mA}} = 37.4 \Omega$$

$$R_{(\text{cut-off})} = \frac{V_{CE(\text{cut-off})}}{I_{CEO}} = \frac{8 \text{ V}}{5 \mu\text{A}} = 1.6 \text{ M}\Omega$$

(d) **Output Voltage**

The output voltage switches between $V_{CE(\text{sat})}$ and V_{CC} as shown below.



◆ 2.13 TRANSISTOR SWITCHING TIMES

Transistors which are designed to operate at high speed between cut-off and saturation are called switching transistors. Figure 2.25 shows the timing characteristics of switching transistor.

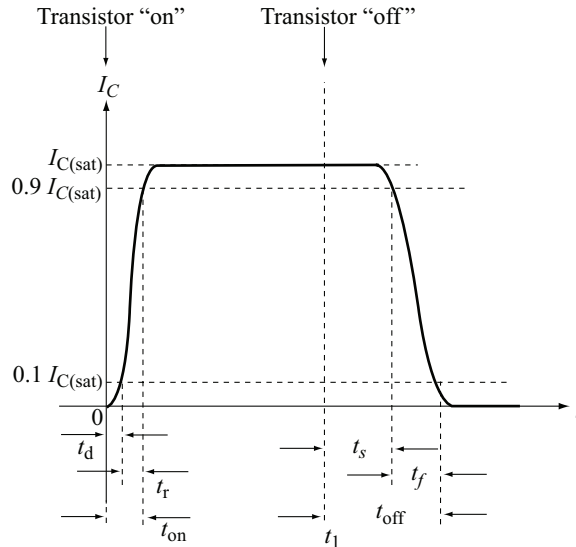


Fig. 2.25 Transistor switching times

At $t = 0$ the transistor switch is turned on by applying a high voltage at the input. The collector current does not rise immediately to $I_{C(sat)}$. Instead it takes a finite turn on time t_{on} to reach the steady state value of $I_{C(sat)}$.

$$t_{on} = t_d + t_r \quad (2.81)$$

t_d = delay time

t_r = rise time

Delay time is the time taken by the collector current to rise from zero to 10% of $I_{C(sat)}$.

Rise time is the time taken by the collector current to rise from 10% to 90% of $I_{C(sat)}$.

At $t = t_1$, the switch is turned off by reducing the input voltage to zero. The transistor does not turn-off immediately. Instead, it takes a finite time t_{off} to turn-off.

$$t_{off} = t_s + t_f \quad (2.82)$$

t_s = storage time

t_f = fall time

Due to stored charges in the base, the collector current remains at $I_{C(sat)}$ over the interval t_s and later it starts decreasing towards zero.

The time taken by the collector current to decrease from 90% to 10% of $I_{C(sat)}$ during turn-off is called the fall time.

Switching transistors are designed with small values of t_{on} and t_{off} in order to operate at higher speeds.

For general purpose transistor 2N4123 the following switching times are available on data sheet at $I_C = 10$ mA.

$$\begin{aligned}
 t_s &= 120 \text{ ns} \\
 t_d &= 25 \text{ ns} \\
 t_r &= 13 \text{ ns} \\
 t_f &= 12 \text{ ns} \\
 \therefore t_{on} &= t_r + t_d = 13 \text{ ns} + 25 \text{ ns} = 38 \text{ ns} \\
 t_{off} &= t_s + t_f = 120 \text{ ns} + 12 \text{ ns} = 132 \text{ ns}
 \end{aligned}$$

where as for the switching transistor BSV52L the data sheet provides:

$$t_{on} = 12 \text{ ns} \quad \text{and} \quad t_{off} = 18 \text{ ns}$$

Note that switching transistors have very small values of t_{on} and t_{off} .

◆ 2.14 BIAS STABILISATION

The collector current in a transistor is sensitive to each of the following parameters

- I_{CO} doubles for every 10°C rise in temperature.
- V_{BE} decreases about 2.5 mV per degree celcius increase in temperature
- β increases with increase in temperature

The three stability factors $S(I_{CO})$, $S(\beta)$ and $S(V_{BE})$ which have been already defined Section 2.4, indicates how I_C changes when I_{CO} , β and V_{BE} changes with temperature. Biasing networks should be designed with low stability factors inorder to obtain higher degree of stability of the operating point. Now let us study the bias stability of various bias configurations.

◆ 2.15 GENERAL EXPRESSION FOR STABILITY FACTOR $S(I_{CO})$

The expression for collector current in terms of β and I_{CO} is given by

$$I_C = (1 + \beta) I_{CO} + \beta I_B \quad (2.83)$$

The stability factors $S(I_{CO})$ is defined as

$$S(I_{CO}) = \left. \frac{\partial I_C}{\partial I_{CO}} \right|_{\beta, V_{BE} \text{ constant}} \quad (2.84)$$

Differentiating Equation (2.83) partially with respect to I_{CO} , treating β constant, we get

$$1 = (1 + \beta) \frac{\partial I_{CO}}{\partial I_C} + \beta \frac{\partial I_B}{\partial I_C}$$

$$(1 + \beta) \frac{\partial I_{CO}}{\partial I_C} = 1 - \beta \frac{\partial I_B}{\partial I_C}$$

$$S(I_{CO}) = \frac{\partial I_C}{\partial I_{CO}} = \frac{1 + \beta}{1 - \beta \frac{\partial I_B}{\partial I_C}} \quad (2.85)$$

Stability factor $S(I_{CO})$ against variation in I_{CO} for any configuration can be found by computing $\frac{\partial I_B}{\partial I_C}$.

Recall that, lower the stability factors, the lower is the variation of I_C with respect to I_{CO} , β and V_{BE} . An observation of Equation (2.85) shows that for $S(I_{CO})$ to be small

$$1 - \beta \frac{\partial I_B}{\partial I_C} \gg 1 + \beta$$

or

$$\frac{\partial I_B}{\partial I_C} \ll -1 \Rightarrow \left| \frac{\partial I_B}{\partial I_C} \right| \gg 1$$

This implies that the variation of $\frac{\partial I_B}{\partial I_C}$ must be negative as well as substantial.

An increase in collector current due to any reason should result in a decrease in the base current.

◆ 2.16 STABILITY FACTORS FOR FIXED-BIAS CIRCUIT

The three stability factors to be determined are $S(I_{CO})$, $S(V_{BE})$ and $S(\beta)$. Other nomenclatures for these stability factors are shown in Table 2.3.

Table 2.3 Other nomenclatures for stability factors

<i>Stability factors</i>		
$S(I_{CO})$	$S(V_{BE})$	$S(\beta)$
S	S'	S''
$S_{I_{CO}}$	$S_{V_{BE}}$	S_{β}

Stability Factor $S(I_{CO})$

$$S(I_{CO}) \equiv \frac{\partial I_C}{\partial I_{CO}} \quad (2.86)$$

From Equation (2.9)

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}$$

$$\begin{aligned} \text{But} \quad & V_{CC} \gg V_{BE} \\ \therefore \quad & I_B \simeq \frac{V_{CC}}{R_B} \quad \text{which is constant} \end{aligned}$$

The analysis of the fixed-bias circuit can be found in Section 2.5.
Differentiating with respect to I_C ,

$$\frac{\partial I_B}{\partial I_C} = 0$$

Substituting in Equation (2.85)

$$S(I_{CO}) = 1 + \beta \quad (2.87)$$

$S(I_{CO})$ is very large because β is usually very large, say, typically that $\beta = 100$ for a transistor. Equation (2.87) implies that $S(I_{CO}) = 101$. This means that $\frac{\Delta I_C}{\Delta I_{CO}} = 101$ or $\Delta I_C = 101 \Delta I_{CO}$. This implies that I_C changes by nearly hundred times the change in I_{CO} , indicating very poor stability of the operating point. This basically happens because in this configuration I_B is independent of I_C .

Stability Factor $S(V_{BE})$

$$S(V_{BE}) \equiv \frac{\partial I_C}{\partial V_{BE}} \quad (2.88)$$

From Equation (2.9)

$$V_{CC} = I_B R_B + V_{BE} \quad (2.89)$$

$$\therefore V_{BE} = V_{CC} - I_B R_B \quad (2.90)$$

$$\text{But} \quad I_B \simeq \frac{I_C}{\beta}$$

Substituting in Equation (2.90)

$$V_{BE} = V_{CC} - \frac{I_C}{\beta} R_B \quad (2.91)$$

Differentiating with respect to V_{BE} , keeping β constant, we get

$$1 = 0 - \frac{R_B}{\beta} \frac{\partial I_C}{\partial V_{BE}} \quad (2.92)$$

From Equation (2.88) and Equation (2.92)

$$S(V_{BE}) = -\frac{\beta}{R_B} \quad (2.93)$$

The negative sign implies that if V_{BE} decreases, due to, say an increase in temperature, I_C will increase. In order to minimize variation of I_C with V_{BE} , the ratio $\frac{\beta}{R_B}$ needs to be kept low either by the choice of a large R_B or by using transistor with small β .

Stability Factor $S(\beta)$

$$S(\beta) \equiv \frac{\partial I_C}{\partial \beta}$$

Since the variation of I_C with β cannot be obtained in analytical form, $S(\beta)$ is computed as

$$S(\beta) = \frac{\Delta I_C}{\Delta \beta} \quad (2.94)$$

From Equation (2.91)

$$I_C = \frac{\beta}{R_B} [V_{CC} - V_{BE}] \quad (2.95)$$

Let $I_C = I_{C_1}$ and $\beta = \beta_1$ at temperature T_1 and

$I_C = I_{C_2}$ and $\beta = \beta_2$ at temperature T_2

$$I_{C_1} = \frac{\beta_1}{R_B} (V_{CC} - V_{BE}) \quad (2.96)$$

$$I_{C_2} = \frac{\beta_2}{R_B} (V_{CC} - V_{BE}) \quad (2.97)$$

Dividing Equation (2.97) by Equation (2.96)

$$\frac{I_{C_2}}{I_{C_1}} = \frac{\beta_2}{\beta_1} \quad (2.98)$$

Subtracting 1 on both sides,

$$\frac{I_{C_2}}{I_{C_1}} - 1 = \frac{\beta_2}{\beta_1} - 1$$

$$\frac{I_{C_2} - I_{C_1}}{I_{C_1}} = \frac{\beta_2 - \beta_1}{\beta_1}$$

$$\frac{\Delta I_C}{I_{C_1}} = \frac{\Delta \beta}{\beta_1}$$

$$\therefore S(\beta) = \frac{\Delta I_C}{\Delta \beta} = \frac{I_{C_1}}{\beta_1} \quad (2.99)$$

From Equation (2.99) it is clear that to keep $S(\beta)$ low, the β of the transistor must be large. This is in contrast to $S(I_{CO})$ and $S(V_{BE})$ which require small values of β to remain low.

◆ 2.17 STABILITY FACTORS FOR EMITTER-BIAS CIRCUIT

Let us now obtain expression $S(I_{CO})$, $S(V_{BE})$ and $S(\beta)$ for the emitter-bias circuit.

Stability Factor $S(I_{CO})$

Refer Fig 2.11

From Equation (2.85), we have

$$S(I_{CO}) = \frac{1 + \beta}{1 - \beta \frac{\partial I_B}{\partial I_C}} \quad (2.100)$$

From Equation (2.15)

$$\begin{aligned} V_{CC} &= I_B R_B + V_{BE} + I_E R_E \\ \text{But } I_E &= I_B + I_C \\ \therefore V_{CC} &= I_B R_B + V_{BE} + I_B R_E + I_C R_E \end{aligned} \quad (2.101)$$

Differentiating Equation (2.101) with respect to I_C keeping V_{BE} constant, we get

$$\begin{aligned} 0 &= \frac{\partial I_B}{\partial I_C} R_B + \frac{\partial I_B}{\partial I_C} R_E + R_E = \frac{\partial I_B}{\partial I_C} (R_B + R_E) + R_E \\ \therefore \frac{\partial I_B}{\partial I_C} &= - \frac{R_E}{R_B + R_E} \end{aligned} \quad (2.102)$$

Substituting Equation (2.102) in Equation (2.100)

$$\begin{aligned} S(I_{CO}) &= \frac{1 + \beta}{1 - \beta \left(\frac{-R_E}{R_B + R_E} \right)} = \frac{1 + \beta}{1 + \beta \left(\frac{R_E}{R_B + R_E} \right)} \\ &= \frac{(\beta + 1)(R_E + R_B)}{R_B + R_E + \beta R_E} = \frac{(\beta + 1)(R_E + R_B)}{(\beta + 1)R_E + R_B} \\ \therefore S(I_{CO}) &= \frac{(\beta + 1) \left(1 + \frac{R_B}{R_E} \right)}{(\beta + 1) + \frac{R_B}{R_E}} \end{aligned} \quad (2.103)$$

Design Considerations

$$\begin{aligned} \text{If } \frac{R_B}{R_E} &\ll (1 + \beta) \\ \text{or } (1 + \beta) R_E &\gg R_B \end{aligned}$$

From Equation (2.103), we get

$$S(I_{CO}) \approx 1 + \frac{(1 + \beta) \left[1 + \frac{R_B}{R_E} \right]}{(1 + \beta)}$$

or
$$S(I_{CO}) \approx 1 + \frac{R_B}{R_E}$$

The smallest achievable value of $S(I_{CO})$ is therefore 1 under the condition $\frac{R_B}{R_E} \ll 1$.

Variation of $S(I_{CO})$ of emitter bias circuit with $\frac{R_B}{R_E}$

For $\frac{R_B}{R_E} \ll 1$, $S(I_{CO}) \approx 1$ as shown in the previous section.

For $\frac{R_B}{R_E} > (1 + \beta)$, from Equation (2.103) we have

$$S(I_{CO}) \approx \frac{(1 + \beta) \left[1 + \frac{R_B}{R_E} \right]}{\frac{R_B}{R_E}} \approx (1 + \beta)$$

For $1 < \frac{R_B}{R_E} < (1 + \beta)$
$$S(I_{CO}) \approx 1 + \frac{R_B}{R_E} \approx \frac{R_B}{R_E}$$

Figure 2.26 shows the variation of $S(I_{CO})$ with $\frac{R_B}{R_E}$.

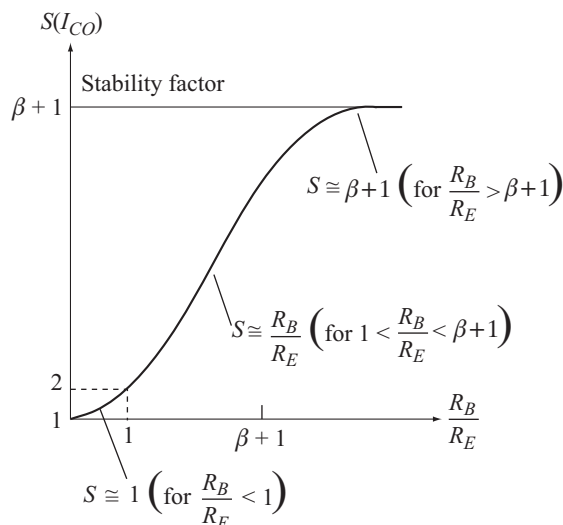


Fig. 2.26 Variation of $S(I_{CO})$ with $\frac{R_B}{R_E}$

(ii) Stability Factor $S(V_{BE})$

By definition

$$S(V_{BE}) = \frac{\partial I_C}{\partial V_{BE}}$$

From Equation (2.101)

$$V_{CC} = I_B R_B + V_{BE} + I_B R_E + I_C R_E$$

Substituting $I_B = \frac{I_C}{\beta}$

$$\begin{aligned} V_{CC} &= \frac{I_C}{\beta} R_B + V_{BE} + \frac{I_C}{\beta} R_E + I_C R_E \\ &= \left[\frac{R_B}{\beta} + \frac{R_E}{\beta} + R_E \right] I_C + V_{BE} \\ &= \left[\frac{R_B + R_E + \beta R_E}{\beta} \right] I_C + V_{BE} \\ V_{CC} &= \left[\frac{R_B + (1 + \beta) R_E}{\beta} \right] I_C + V_{BE} \end{aligned} \quad (2.104)$$

Differentiating Equation (2.104) with respect to I_C , keeping β constant, we get

$$\begin{aligned} 0 &= \frac{R_B + (1 + \beta) R_E}{\beta} + \frac{\partial V_{BE}}{\partial I_C} \\ \therefore S(V_{BE}) &= \frac{\partial I_C}{\partial V_{BE}} = \frac{-\beta}{R_B + (1 + \beta) R_E} \end{aligned} \quad (2.105)$$

Design Considerations

If $(1 + \beta) R_E \gg R_B$

From Equation (2.105) we have

$$S(V_{BE}) \approx \frac{-\beta}{(1 + \beta) R_E}$$

For large β , $(1 + \beta) \approx \beta$

$$S(V_{BE}) \approx \frac{-1}{R_E}$$

Note that $S(V_{BE})$ can be made small by increasing the value of R_E .

(iii) Stability Factor $S(\beta)$

By definition

$$S(\beta) = \frac{\partial I_C}{\partial \beta}$$

But to find $S(\beta)$ we use

$$S(\beta) = \frac{\Delta I_C}{\Delta \beta}$$

From Equation (2.104)

$$\begin{aligned} V_{CC} &= \left[\frac{R_B + (1 + \beta)R_E}{\beta} \right] I_C + V_{BE} \\ \therefore I_C &= \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta)R_E} \beta \end{aligned} \quad (2.106)$$

Let $I_C = I_{C_1}$ and $\beta = \beta_1$ at temperature T_1 and $I_C = I_{C_2}$ and $\beta = \beta_2$ at temperature T_2 . Substituting in Equation (2.106) we obtain

$$I_{C_1} = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta_1)R_E} \beta_1 \quad (2.107)$$

$$I_{C_2} = \frac{V_{CC} - V_{BE}}{R_B + (1 + \beta_2)R_E} \beta_2 \quad (2.108)$$

Dividing Equation (2.108) by Equation (2.107)

$$\begin{aligned} \frac{I_{C_2}}{I_{C_1}} &= \frac{R_B + (\beta_1 + 1)R_E}{R_B + (\beta_2 + 1)R_E} \frac{\beta_2}{\beta_1} \\ \frac{I_{C_2}}{I_{C_1}} &= \frac{1 + \beta_1 + \frac{R_B}{R_E}}{1 + \beta_2 + \frac{R_B}{R_E}} \frac{\beta_2}{\beta_1} \end{aligned} \quad (2.109)$$

Subtracting 1 from both sides of Equation (2.109)

$$\begin{aligned} \frac{I_{C_2}}{I_{C_1}} - 1 &= \frac{1 + \beta_1 + \frac{R_B}{R_E}}{1 + \beta_2 + \frac{R_B}{R_E}} \frac{\beta_2}{\beta_1} - 1 \\ \frac{I_{C_2} - I_{C_1}}{I_{C_1}} &= \frac{\left(1 + \beta_1 + \frac{R_B}{R_E}\right) \beta_2 - \left(1 + \beta_2 + \frac{R_B}{R_E}\right) \beta_1}{\beta_1 \left(1 + \beta_2 + \frac{R_B}{R_E}\right)} \\ \therefore \frac{\Delta I_C}{I_{C_1}} &= \frac{\beta_2 + \beta_1 \beta_2 + \frac{R_B}{R_E} \beta_2 - \beta_1 - \beta_1 \beta_2 - \frac{R_B}{R_E} \beta_1}{\beta_1 \left(1 + \beta_2 + \frac{R_B}{R_E}\right)} \end{aligned}$$

$$\begin{aligned} \frac{\Delta I_C}{I_{C_1}} &= \frac{(\beta_2 - \beta_1) + \frac{R_B}{R_E}(\beta_2 - \beta_1)}{\beta_1 \left(1 + \beta_2 + \frac{R_B}{R_E}\right)} = \frac{(\beta_2 - \beta_1) \left(1 + \frac{R_B}{R_E}\right)}{\beta_1 \left(1 + \beta_2 + \frac{R_B}{R_E}\right)} \\ \therefore S(\beta) &= \frac{\Delta I_C}{\Delta \beta} = \frac{I_{C_1} \left(1 + \frac{R_B}{R_E}\right)}{\beta_1 \left(1 + \beta_2 + \frac{R_B}{R_E}\right)} \end{aligned} \quad (2.110)$$

Design Considerations

$$(1 + \beta_2) \approx \beta_2$$

From Equation (2.110) we have

$$S(\beta) \approx \frac{I_{C_1} \left[1 + \frac{R_B}{R_E}\right]}{\beta_1 \left[\beta_2 + \frac{R_B}{R_E}\right]} \approx \frac{I_{C_1} \left[1 + \frac{R_B}{R_E}\right]}{\beta_1 \beta_2 \left[1 + \frac{R_B}{\beta_2 R_E}\right]}$$

For large β , $\frac{R_B}{\beta_2 R_E} \ll 1$

$$\therefore 1 + \frac{R_B}{\beta_2 R_E} \approx 1$$

$$\therefore S(\beta) \approx \frac{I_{C_1}}{\beta_1 \beta_2} \left[1 + \frac{R_B}{R_E}\right] = \frac{I_{C_1}}{\beta_1 \beta_2} S(I_{CO})$$

Thus $S(\beta)$ can be minimized by selecting a transistor with large β as well as by keeping $\frac{R_B}{R_E} \ll 1$.

◆ 2.18 STABILITY FACTORS FOR VOLTAGE DIVIDER BIAS CIRCUIT

Let us proceed on the same lines as in the case of other configurations.

Refer Fig. 2.16.

(i) *Stability Factor $S(I_{CO})$*

$$S(I_{CO}) = \frac{1 + \beta}{1 - \beta \frac{\partial I_B}{\partial I_C}}$$

From Equation (2.29)

$$\begin{aligned} V_{Th} &= R_{Th} I_B + V_{BE} + I_E R_E \\ \text{But } I_E &= I_C + I_B \\ V_{Th} &= R_{Th} I_B + V_{BE} + I_C R_E + I_B R_E \end{aligned} \quad (2.111)$$

Differentiating the above equation partially with respect to I_C , keeping V_{BE} constant, we get

$$\begin{aligned} 0 &= (R_E + R_{Th}) \frac{\partial I_B}{\partial I_C} + R_E \\ \therefore \frac{\partial I_B}{\partial I_C} &= \frac{-R_E}{R_E + R_{Th}} \end{aligned}$$

Substituting in the expression for $S(I_{CO})$, we get

$$\begin{aligned} S(I_{CO}) &= \frac{1+\beta}{1+\beta \frac{R_E}{R_E + R_{Th}}} = \frac{1+\beta}{R_E + R_{Th} + \beta R_E} (R_E + R_{Th}) \\ \therefore S(I_{CO}) &= (\beta + 1) \frac{R_E + R_{Th}}{(\beta + 1)R_E + R_{Th}} \end{aligned} \quad (2.112)$$

$$\therefore S(I_{CO}) = (\beta + 1) \frac{\left[1 + \frac{R_{Th}}{R_E} \right]}{(\beta + 1) + \frac{R_{Th}}{R_E}} \quad (2.113)$$

Design Considerations

Consider the denominator of Equation (2.112). Generally $(\beta + 1) R_E \gg R_{Th}$

$$\text{Hence, } S(I_{CO}) \simeq \frac{(\beta + 1)(R_E + R_{Th})}{(\beta + 1)R_E} = 1 + \frac{R_{Th}}{R_E}$$

The smallest achievable value of $S(I_{CO})$ is therefore 1 under the condition $R_{Th} \ll R_E$.

(ii) Stability Factor $S(V_{BE})$

By definition

$$S(V_{BE}) = \frac{\partial I_C}{\partial V_{BE}}$$

From Equation (2.111)

$$V_{Th} = R_{Th} I_B + V_{BE} + I_C R_E + I_B R_E$$

Substituting

$$I_B = \frac{I_C}{\beta}$$

$$\begin{aligned}
\therefore V_{Th} &= \left(\frac{R_{Th}}{\beta} + \frac{R_E}{\beta} + R_E \right) I_C + V_{BE} \\
&= \left(\frac{R_{Th} + R_E + \beta R_E}{\beta} \right) I_C + V_{BE} \\
V_{Th} &= \left(\frac{(R_{Th} + (\beta + 1)R_E)}{\beta} \right) I_C + V_{BE} \quad (2.114)
\end{aligned}$$

Differentiating partially with respect to V_{BE} , keeping β constant

$$\begin{aligned}
0 &= \frac{R_{Th} + (\beta + 1)R_E}{\beta} \frac{\partial I_C}{\partial V_{BE}} + 1 \\
\therefore S(V_{BE}) &= \frac{\partial I_C}{\partial V_{BE}} = \frac{-\beta}{R_{Th} + (\beta + 1)R_E} \quad (2.115)
\end{aligned}$$

Design Considerations

Again from Equation (2.115)

If $(\beta + 1) R_E \gg R_{Th}$,

$$S(V_{BE}) \simeq \frac{-\beta}{(\beta + 1)R_E}$$

For large β , $\beta + 1 \simeq \beta$

$$\therefore S(V_{BE}) = -\frac{1}{R_E}$$

Thus, $S(V_{BE})$ can be made small by increasing the value of emitter resistance R_E .

(iii) Stability Factor $S(\beta)$

By definition

$$S(\beta) = \frac{\Delta I_C}{\Delta \beta}$$

From Equation (2.114)

$$\begin{aligned}
V_{Th} &= \left(\frac{R_{Th} + (\beta + 1)R_E}{\beta} \right) I_C + V_{BE} \\
\therefore \frac{\partial R_{Th} + (\beta + 1)R_E}{\partial \beta} I_C &= V_{Th} - V_{BE} \\
I_C &= \frac{(V_{Th} - V_{BE})\beta}{R_{Th} + (1 + \beta)R_E}
\end{aligned}$$

Let $I_C = I_{C_1}$ and $\beta = \beta_1$ at temperature T_1 and

Let $I_C = I_{C_2}$ and $\beta = \beta_2$ at temperature T_2

$$\therefore I_{C_1} = \frac{(V_{Th} - V_{BE})\beta_1}{R_{Th} + (1 + \beta_1)R_E} \quad (2.116)$$

$$\text{and } I_{C_2} = \frac{(V_{Th} - V_{BE})\beta_2}{R_{Th} + (1 + \beta_2)R_E} \quad (2.117)$$

Dividing Equation (2.117) by Equation (2.116)

$$\frac{I_{C_2}}{I_{C_1}} = \frac{R_{Th} + (1 + \beta_1)R_E}{R_{Th} + (1 + \beta_2)R_E} \cdot \frac{\beta_2}{\beta_1} \quad (2.118)$$

Subtracting 1 from both sides of Equation (2.118)

$$\begin{aligned} \frac{I_{C_2}}{I_{C_1}} - 1 &= \frac{R_{Th} + (1 + \beta_1)R_E}{R_{Th} + (1 + \beta_2)R_E} \cdot \frac{\beta_2}{\beta_1} - 1 \\ \frac{DI_C}{I_{C_1}} &= \frac{[R_{Th} + (1 + \beta_1)R_E] \beta_2 - [R_{Th} + (1 + \beta_2)R_E] \beta_1}{[R_{Th} + (1 + \beta_2)R_E] \beta_1} \\ &= \frac{R_{Th} \beta_2 + \beta_2 R_E + \beta_1 \beta_2 R_E - R_{Th} \beta_1 - \beta_1 R_E - \beta_1 \beta_2 R_E}{[R_{Th} + (1 + \beta_2)R_E] \beta_1} \\ &= \frac{R_{Th} (\beta_2 - \beta_1) + R_E (\beta_2 - \beta_1)}{[R_{Th} + (1 + \beta_2)R_E] \beta_1} \\ \therefore \frac{DI_C}{D\beta} &= \frac{I_{C_1}}{\beta_1} \cdot \frac{R_{Th} + R_E}{R_{Th} + (1 + \beta_2)R_E} \end{aligned} \quad (2.119)$$

$$\therefore S(\beta) = \frac{DI_C}{D\beta} = \frac{I_{C_1} \left(\frac{R_{Th}}{R_E} + 1 \right)}{\beta_1 (\beta_2 + 1) \left(\frac{R_{Th}}{R_E} + \beta_2 + 1 \right)} \quad (2.120)$$

Design Considerations

From Equation (2.120)

$$S(\beta) = \frac{I_{C_1} \left(\frac{R_{Th}}{R_E} + 1 \right)}{\beta_1 (\beta_2 + 1) \left(\frac{R_{Th}}{R_E} + \beta_2 + 1 \right)}$$

Now

$$\beta_2 + 1 \simeq \beta_2$$

$$\therefore S(\beta) = \frac{I_{C_1} \left(1 + \frac{R_{Th}}{R_E} \right)}{\beta_1 \beta_2 \left(1 + \frac{R_{Th}}{\beta_2 R_E} \right)} \quad (2.121)$$

For large β $\frac{R_{Th}}{\beta_2 R_E} \ll 1$

$$\therefore 1 + \frac{R_{Th}}{\beta_2 R_E} \simeq 1$$

From Equation (2.121)

$$\therefore S(\beta) = \frac{I_{C_1} \left(1 + \frac{R_{Th}}{R_E} \right)}{\beta_1 \beta_2} = \frac{I_{C_1}}{\beta_1 \beta_2} S(I_{CO})$$

Thus, $S(\beta)$ can be minimized by selecting a transistor with large β as well as by keeping $\frac{R_{Th}}{R_E} \ll 1$.

◆ 2.19 STABILITY FACTORS FOR COLLECTOR FEEDBACK BIAS CIRCUIT

Let us proceed on the same lines as in the case of other configurations. Refer Fig. 2.19.

(i) *Stability factor $S(I_{CO})$*

$$S(I_{CO}) = \frac{1 + \beta}{1 - \beta \frac{\partial I_B}{\partial I_C}}$$

From Equation (2.47)

$$\begin{aligned} V_{CC} &= (I_C + I_B) R_C + I_B R_B + V_{BE} + I_E R_E \\ &= (I_C + I_B) R_C + I_B R_B + V_{BE} + (I_B + I_C) R_E \\ V_{CC} &= (R_C + R_E) I_C + (R_C + R_E + R_B) I_B + V_{BE} \end{aligned} \quad (2.122)$$

Differentiating Equation (2.122) partially with respect to I_C ,

$$0 = (R_C + R_E) + (R_C + R_E + R_B) \frac{\partial I_B}{\partial I_C}$$

$$\therefore \frac{\partial I_B}{\partial I_C} = - \frac{R_C + R_E}{R_C + R_E + R_B}$$

Now
$$S(I_{CO}) = - \frac{1 + \beta}{1 - \beta \frac{\partial I_B}{\partial I_C} (R_C + R_E + R_B)}$$

$$S(I_{CO}) = \frac{(\beta + 1)(R_C + R_E + R_B)}{R_B + (1 + \beta)(R_C + R_E)} \quad (2.123)$$

$$\therefore S(I_{CO}) = \frac{(\beta + 1) + \frac{R_B}{R_C + R_E}}{(1 + \beta) + \frac{R_B}{R_C + R_E}} \quad (2.124)$$

(ii) Stability Factor $S(V_{BE})$

By definition
$$S(V_{BE}) = \frac{\partial I_C}{\partial V_{BE}}$$

From Equation (2.122)

$$\begin{aligned} V_{CC} &= (R_C + R_E) I_C + (R_C + R_E + R_B) I_B + V_{BE} \\ V_{CC} &= (R_C + R_E) I_C + (R_C + R_E + R_B) \frac{I_C}{\beta} + V_{BE} \end{aligned} \quad (2.125)$$

Differentiating Equation (2.125) partially with respect to I_C

$$\begin{aligned} 0 &= (R_C + R_E) + \frac{(R_C + R_E + R_B)}{\beta} + \frac{\partial V_{BE}}{\partial I_C} \\ 0 &= \frac{\beta(R_C + R_E) + (R_C + R_E + R_B)}{\beta} + \frac{\partial V_{BE}}{\partial I_C} \\ \therefore S(V_{BE}) &= \frac{\partial I_C}{\partial V_{BE}} = \frac{-\beta}{\beta(R_C + R_E) + (R_C + R_E + R_B)} \\ S(V_{BE}) &= \frac{-\beta}{(\beta + 1)(R_C + R_E) + R_B} \end{aligned} \quad (2.126)$$

(ii) Stability Factor $S(\beta)$

By definition
$$S(\beta) = \frac{\partial I_C}{\partial \beta}$$

From Equation (2.125)

$$I_C = \frac{\beta(V_{CC} - V_{BE})}{R_B + (1 + \beta)(R_C + R_E)}$$

Let $I_C = I_{C_1}$ and $\beta = \beta_1$ at temperature T_1 and

$I_C = I_{C_2}$ and $\beta = \beta_2$ at temperature T_2

$$\therefore I_{C_1} = \frac{\beta_1(V_{CC} - V_{BE})}{R_B + (1 + \beta_1)(R_C + R_E)} \quad (2.127)$$

$$I_{C_2} = \frac{\beta_2(V_{CC} - V_{BE})}{R_B + (1 + \beta_2)(R_C + R_E)} \quad (2.128)$$

Dividing Equation (2.128) by Equation (2.127), we get

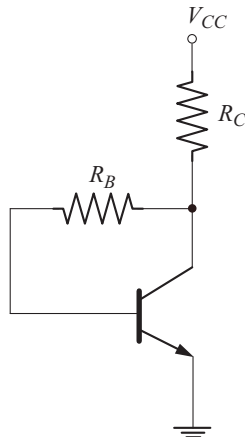
$$\frac{I_{C_2}}{I_{C_1}} = \frac{R_B + (1 + \beta_1)(R_C + R_E)}{R_B + (1 + \beta_2)(R_C + R_E)} \cdot \frac{\beta_2}{\beta_1} \quad (2.129)$$

Subtracting 1 from both sides of Equation (2.129)

$$\begin{aligned} \frac{I_{C_2}}{I_{C_1}} - 1 &= \frac{R_B + (1 + \beta_1)(R_C + R_E)}{R_B + (1 + \beta_2)(R_C + R_E)} \cdot \frac{\beta_2}{\beta_1} - 1 \\ \frac{DI_C}{I_{C_1}} &= \frac{[R_B + (1 + \beta_1)(R_C + R_E)] \beta_2 - [R_B + (1 + \beta_2)(R_C + R_E)] \beta_1}{[R_B + (1 + \beta_2)(R_C + R_E)] \beta_1} \\ &= \frac{R_B \beta_2 + (1 + \beta_1) \beta_2 (R_C + R_E) - R_B \beta_1 - \beta_1 (1 + \beta_2)(R_C + R_E)}{[R_B + (1 + \beta_2)(R_C + R_E)] \beta_1} \\ &= \frac{R_B (\beta_2 - \beta_1) + (R_C + R_E) (\beta_2 - \beta_1)}{[R_B + (1 + \beta_2)(R_C + R_E)] \beta_1} \\ \frac{DI_C}{I_{C_1}} &= \frac{[R_B + (R_C + R_E)] (\beta_2 - \beta_1)}{[R_B + (1 + \beta_2)(R_C + R_E)] \beta_1} \\ \therefore S(\beta) &= \frac{DI_C}{D\beta} = \frac{I_{C_1} [R_B + (R_C + R_E)]}{\beta_1 [R_B + (1 + \beta_2)(R_C + R_E)]} \quad (2.130) \end{aligned}$$

Example 2.39

For the circuit shown below derive the expressions for $S(I_{CO})$, $S(V_{BE})$ and $S(\beta)$.



Solution

The given circuit is nothing but the collector feedback configuration with $R_E = 0 \Omega$.

The required results can be obtained by substituting $R_E = 0 \Omega$ in the results of section 2.19.

From Equation (2.124)

$$S(I_{CO}) = \frac{(1+\beta) \frac{R_B}{R_C} + 1}{(1+\beta) + \frac{R_B}{R_C}}$$

From Equation (2.126)

$$S(V_{BE}) = \frac{-\beta}{R_B + (1+\beta)R_C}$$

From Equation (2.130)

$$S(\beta) = \frac{I_{C1} [R_B + R_C]}{\beta_1 [R_B + (1+\beta_2)R_C]}$$

Example 2.40

Calculate the stability factor $S(I_{CO})$ and the change in I_C from 25°C to 100°C for the transistor which has $I_{CO}(25^\circ \text{C}) = 0.1 \text{ nA}$ and $I_{CO}(100^\circ \text{C}) = 20 \text{ nA}$, for the following emitter bias arrangements. Take $\beta = 50$.

(a) $\frac{R_B}{R_E} = 100$ (b) $\frac{R_B}{R_E} = 0.1$

Also calculate I_{CQ} at 100°C for each case if I_{CQ} at 25°C is 2 mA .

Solution

For the emitter bias configuration

$$S(I_{CO}) = (1+\beta) \frac{1 + \frac{R_B}{R_E}}{(1+\beta) + \frac{R_B}{R_E}}$$

(a) $\frac{R_B}{R_E} = 100$

$$S(I_{CO}) = \frac{(51)(1+100)}{51+100} = 34.11$$

$$\Delta I_C = S(I_{CO}) \Delta I_{CO}$$

$$\begin{aligned} \Delta I_{CO} &= I_{CO}(100^\circ \text{C}) - I_{CO}(25^\circ \text{C}) \\ &= 20 \text{ nA} - 0.1 \text{ nA} = 19.9 \text{ nA} \end{aligned}$$

$$\Delta I_C = (34.11) (19.9 \text{ nA})$$

$$= 0.67 \mu\text{A}$$

$$I_{C_2} = I_{CQ} \text{ at } 100^\circ \text{ C}$$

$$I_{C_1} = I_{CQ} \text{ at } 25^\circ \text{ C} = 2 \text{ mA}$$

$$I_{C_2} = I_{C_1} + \Delta I_C$$

$$= 2 \text{ mA} + 0.67 \mu\text{A}$$

$$\approx 2 \text{ mA}$$

$$(b) \frac{R_B}{R_E} = 0.1$$

$$S(I_{CO}) = \frac{(51)(1.1)}{51 + 0.1} = 1.09$$

$$\Delta I_C = S(I_{CO}) \Delta I_{CO}$$

$$= (1.09) (19.9 \text{ nA}) = 21.69 \text{ nA}$$

$$I_{C_2} = I_{C_1} + \Delta I_C$$

$$= 2 \text{ mA} + 21.69 \text{ nA}$$

$$\approx 2 \text{ mA}$$

Observe that changes I_C is minimal when $\frac{R_B}{R_E} \ll 1$.

Example 2.41

Determine the stability factor $S(V_{BE})$ and the change in I_C from 25° C to 100° C for the transistor with $V_{BE}(25^\circ \text{ C}) = 0.65 \text{ V}$ and $V_{BE}(100^\circ \text{ C}) = 0.48 \text{ V}$ for the following bias arrangements

(a) Fixed bias with $R_B = 270 \text{ k}\Omega$ and $\beta = 120$.

(b) Emitter bias with $R_B = 39 \text{ k}\Omega$, $R_E = 1.2 \text{ k}\Omega$ and $\beta = 120$.

(c) Voltage divider bias with $R_1 = 39 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, $R_E = 1 \text{ k}\Omega$ and $\beta = 120$.

Solution

$$\Delta V_{BE} = V_{BE}(100^\circ \text{ C}) - V_{BE}(25^\circ \text{ C})$$

$$= 0.48 \text{ V} - 0.65 \text{ V} = -0.17 \text{ V}$$

(a) **Fixed Bias**

$$R_B = 270 \text{ k}\Omega \quad \beta = 120$$

$$S(V_{BE}) = \frac{-\beta}{R_B} = -\frac{120}{270 \text{ k}\Omega} = -0.44 \times 10^{-3}$$

$$\Delta I_C = S(V_{BE}) \Delta V_{BE}$$

$$= (-0.44 \times 10^{-3}) (-0.17) = 74.8 \mu\text{A}$$

(b) **Emitter Bias**

$$R_B = 39 \text{ k}\Omega \quad R_E = 1.2 \text{ k}\Omega \quad \beta = 120.$$

$$\begin{aligned} S(V_{BE}) &= \frac{-\beta}{R_B + (1+\beta)R_E} \\ &= \frac{-120}{39 \text{ k}\Omega + (121)(1.2 \text{ k}\Omega)} = -0.65 \times 10^{-3} \\ \Delta I_C &= S(V_{BE}) \Delta V_{BE} = (-0.65 \times 10^{-3})(-0.17) = 110.5 \text{ }\mu\text{A} \end{aligned}$$

(c) **Voltage Divider Bias**

$$R_1 = 39 \text{ k}\Omega \quad R_2 = 10 \text{ k}\Omega \quad R_E = 1 \text{ k}\Omega \quad \beta = 120$$

$$\begin{aligned} S(V_{BE}) &= \frac{-\beta}{R_{Th} + (1+\beta)R_E} \\ R_{th} &= R_1 \parallel R_2 = 39 \text{ k}\Omega \parallel 10 \text{ k}\Omega = 7.95 \text{ k}\Omega \\ S(V_{BE}) &= \frac{-120}{7.95 \text{ k}\Omega + (121)(1 \text{ k}\Omega)} = -0.93 \times 10^{-3} \\ \Delta I_C &= S(V_{BE}) \Delta V_{BE} = (-0.93 \times 10^{-3})(-0.17) = 158.1 \text{ }\mu\text{A} \end{aligned}$$

Example 2.42

Determine the stability factor $S(\beta)$ and the change in I_C from 25°C to 100°C for the transistor with $\beta(25^\circ \text{C}) = 50$ and $\beta(100^\circ \text{C}) = 100$ for the following bias arrangement.

- (a) Fixed bias with $R_B = 330 \text{ k}\Omega$
 (b) Emitter bias with $\frac{R_B}{R_E} = 5$
 (c) Voltage divider bias with $\frac{R_{Th}}{R_E} = 1.5$.

Also calculate I_{CQ} at the 100°C in each case if I_{CQ} at 25°C is 3 mA .

Solution

$$\begin{aligned} \beta_1 &= \beta(25^\circ \text{C}) = 50 \\ \beta_2 &= \beta(100^\circ \text{C}) = 100 \\ \Delta \beta &= \beta_2 - \beta_1 = 50 \\ I_{C_1} &= I_{CQ}(25^\circ \text{C}) = 3 \text{ mA} \\ I_{C_2} &= I_{CQ}(100^\circ \text{C}) \end{aligned}$$

(a) **Fixed bias**

$$S(\beta) = \frac{I_{C_1}}{\beta_1} = \frac{3 \times 10^{-3}}{50} = 0.06 \times 10^{-3}$$

$$\Delta I_C = S(\beta) \Delta\beta = (0.06 \times 10^{-3}) (50) = 3 \text{ mA}$$

$$I_{C_2} = I_{C_1} + \Delta I_C = 3 \text{ mA} + 3 \text{ mA} = 6 \text{ mA}$$

Observe the doubling of I_C value when β is doubled.

(b) **Emitter Bias**

$$\frac{R_B}{R_E} = 5$$

$$S(\beta) = \frac{I_{C_1} \left[1 + \frac{R_B}{R_E} \right]}{\beta_1 \left[1 + \beta_2 + \frac{R_B}{R_E} \right]} = \frac{(3 \text{ mA})[1+5]}{50[101+5]} = 3.39 \times 10^{-6}$$

$$\Delta I_C = S(\beta) \Delta\beta = (3.39 \times 10^{-6}) (50) = 0.17 \text{ mA}$$

$$I_{C_2} = I_{C_1} + \Delta I_C = 3 \text{ mA} + 0.17 \text{ mA} = 3.17 \text{ mA}$$

(c) **Voltage Divider Bias**

$$\frac{R_{Th}}{R_E} = 1.5$$

$$S(\beta) = \frac{I_{C_1} \left[1 + \frac{R_{Th}}{R_E} \right]}{\beta_1 \left[1 + \beta_2 + \frac{R_{Th}}{R_E} \right]} = \frac{(3 \text{ mA})[1+1.5]}{50[101+1.5]} = 1.46 \times 10^{-6}$$

$$\Delta I_C = S(\beta) \Delta\beta = (1.46 \times 10^{-6}) (50) = 73 \text{ } \mu\text{A}$$

$$I_{C_2} = I_{C_1} + \Delta I_C = 3 \text{ mA} + 73 \text{ } \mu\text{A} = 3.07 \text{ mA}$$

Example 2.43

For a voltage divider bias circuit, the following values are given :

$$R_1 = 62 \text{ k}\Omega \quad R_2 = 9.1 \text{ k}\Omega \quad R_C = 3.9 \text{ k}\Omega \quad R_E = 680 \text{ } \Omega \quad V_{BE} = 0.7 \text{ V} \quad \beta = 80 \quad V_{CC} = 16 \text{ V}$$

Calculate

(a) $S(I_{CO})$

(b) $S(V_{BE})$

(c) $S(\beta)$ with $\beta(T_1) = 80$ and $\beta(T_2) = 25\%$ more than $\beta(T_1)$

(d) Net change in I_C if I_{CO} increases from 0.2 to 10 μA , V_{BE} drops from 0.7 to 0.5 V and β increases by 25%.

Solution

(a) $S(I_{CO})$

$$S(I_{CO}) = (\beta + 1) \frac{R_E + R_{Th}}{(\beta + 1)R_E + R_{Th}}$$

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2} = \frac{62 \text{ kW} \times 9.1 \text{ kW}}{62 \text{ kW} + 9.1 \text{ kW}} = 7.94 \text{ k}\Omega$$

$$\therefore S(I_{CO}) = \frac{(80+1)(0.68 \text{ kW} + 7.94 \text{ kW})}{(80+1)(0.68 \text{ kW}) + 7.94 \text{ kW}} = 11.08$$

(b) $S(V_{BE})$

$$\begin{aligned} S(V_{BE}) &= \frac{-\beta}{R_{Th} + (\beta+1)R_E} \\ &= \frac{-80}{7.94 \times 10^3 + (80+1)680} = 1.27 \times 10^{-3} \end{aligned}$$

(c) $S(\beta)$

$$S(\beta) = \frac{I_{C1} \frac{\beta_1}{\beta_1 + 1} + \frac{R_{Th}}{R_E} \frac{\beta_1}{\beta_1 + 1}}{\beta_1 \frac{\beta_1}{\beta_1 + 1} + \beta_2 + \frac{R_{Th}}{R_E} \frac{\beta_1}{\beta_1 + 1}}$$

$$I_{C1} = \frac{(V_{Th} - V_{BE}) \beta_1}{R_{Th} + (1 + \beta_1) R_E}$$

$$V_{th} = V_{CC} \frac{R_2}{R_1 + R_2} = (16 \text{ V}) \frac{9.1 \text{ kW}}{62 \text{ kW} + 9.1 \text{ kW}} = 2.05 \text{ V}$$

$$I_{C1} = \frac{(2.05 \text{ V} - 0.7 \text{ V}) \times 80}{7.94 \text{ k}\Omega + (1 + 80)0.68 \text{ k}\Omega} = 1.71 \text{ mA}$$

$$\beta_2 = 25\% \text{ more than } \beta_1 = 1.25 \times 80 = 100$$

$$S(\beta) = \frac{[1.71 \times 10^{-3}] \left[1 + \frac{7.94 \text{ k}\Omega}{0.68 \text{ k}\Omega} \right]}{80 \left[1 + 100 + \frac{7.94 \text{ k}\Omega}{0.68 \text{ k}\Omega} \right]} = 2.41 \times 10^{-6}$$

(d) Net change in collector current, ΔI_C

$$\Delta I_C = S(I_{CO}) \Delta I_{CO} + S(V_{BE}) \Delta V_{BE} + S(\beta) \Delta \beta$$

$$\Delta I_{CO} = 10 \text{ }\mu\text{A} - 0.2 \text{ }\mu\text{A} = 9.8 \text{ }\mu\text{A}$$

$$\Delta V_{BE} = 0.5 \text{ V} - 0.7 \text{ V} = -0.2 \text{ V}$$

$$\Delta \beta = \beta_2 - \beta_1 = 100 - 80 = 20$$

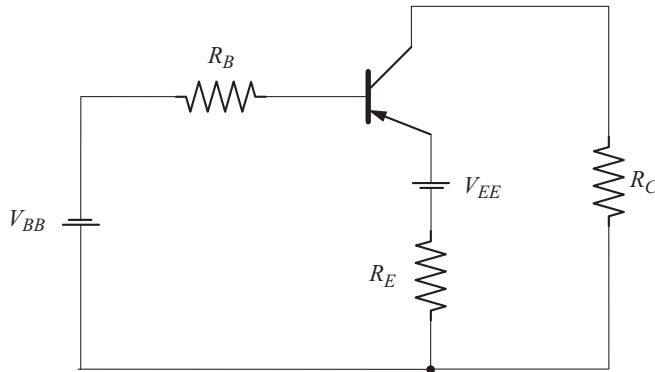
$$\begin{aligned}\Delta I_C &= [11.08] [9.8 \times 10^{-6}] + [-1.27 \times 10^{-3}] [-0.2] \\ &\quad + [2.141 \times 10^{-6}] [20] \\ &= 0.41 \text{ mA}\end{aligned}$$

$$I_{C2} = I_{C1} + \Delta I_C = 1.71 \text{ mA} + 0.41 \text{ mA} = 2.12 \text{ mA}$$

Note: Similarly we can calculate the net change in collector current for the other biasing configurations.

Example 2.44

Find an expression for $S(I_{CO})$ for the circuit shown below.



Solution

The given circuit with actual current directions voltage polarities is redrawn in Fig. A.

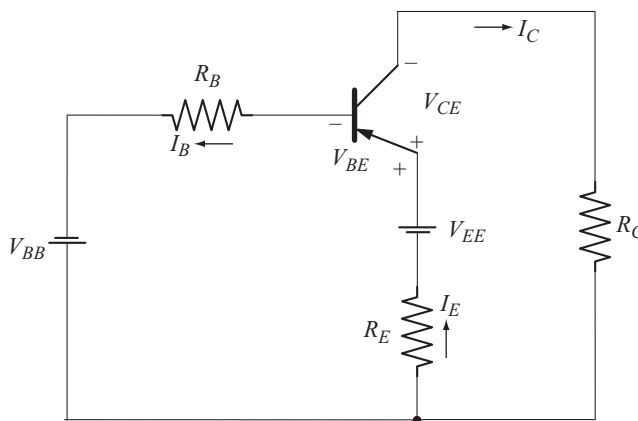


Fig. A

$$S(I_{CO}) = \frac{1 + \beta}{1 - \beta \frac{I_B}{I_C}}$$

Writing the KVL equation to the base-emitter circuit,

$$-V_{BB} + I_B R_B + V_{BE} - V_{EE} + I_E R_E = 0$$

$$\therefore I_B R_B + I_E R_E = V_{BB} + V_{EE} - V_{BE}$$

$$\text{But } I_E = I_B + I_C$$

$$\therefore I_B(R_B + R_E) + I_C R_E = V_{BB} + V_{EE} - V_{BE}$$

Differentiating partially with respect to I_C

$$\frac{\partial I_B}{\partial I_C} (R_B + R_E) + R_E = 0$$

$$\therefore \frac{\partial I_B}{\partial I_C} = \frac{-R_E}{(R_B + R_E)}$$

$$\therefore S(I_{CO}) = \frac{1 + \beta}{1 + \beta \frac{R_E}{R_B + R_E}}$$

$$S(I_{CO}) = \frac{1 + \beta}{\frac{R_B + R_E + \beta R_E}{R_B + R_E}}$$

$$= \frac{(1 + \beta)(R_B + R_E)}{R_B + (1 + \beta)R_E}$$

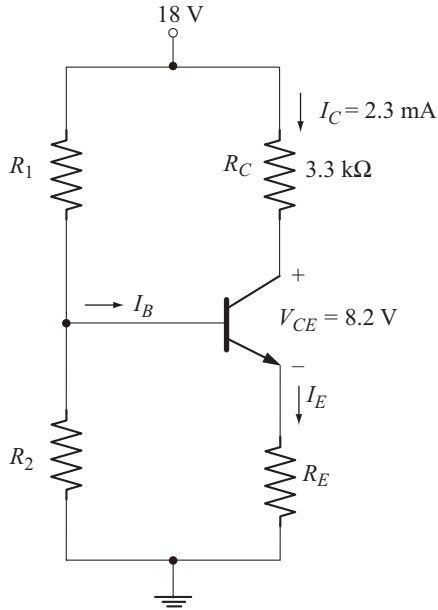
$$S(I_{CO}) = \frac{(1 + \beta) \left(1 + \frac{R_B}{R_E} \right)}{(1 + \beta) + \frac{R_B}{R_E}}$$

Example 2.45

Design a voltage divider bias circuit using a silicon transistor with $V_{CC} = 18 \text{ V}$, $I_C = 2.3 \text{ mA}$, $V_{CE} = 8.2 \text{ V}$, $R_C = 3.3 \text{ k}\Omega$, $\beta = 100$ and $S(I_{CO}) \leq 5$.

Solution

The circuit is shown below.



Given $I_C = 2.3 \text{ mA}$,

$$I_B = \frac{I_C}{\beta} = \frac{2.3 \text{ mA}}{100} = 0.023 \text{ mA}$$

$$I_E = I_B + I_C = 2.3 + 0.023 = 2.32 \text{ mA}$$

$$V_{BE} = 0.7 \text{ V} \quad \text{for a silicon transistor}$$

$$V_{CE} = V_{CC} - I_C (R_C + R_E)$$

$$\therefore R_C + R_E = \frac{V_{CC} - V_{CE}}{I_C}$$

$$\text{and} \quad R_E = \frac{18 \text{ V} - 8.2 \text{ V}}{2.3 \text{ mA}} - 3.3 \text{ k}\Omega = 0.96 \text{ k}\Omega$$

$$S(I_{CO}) = (\beta + 1) \frac{R_E + R_{Th}}{(\beta + 1) R_E + R_{Th}}$$

$$S(I_{CO}) \leq 5 \text{ implies}$$

$$(\beta + 1) \frac{R_E + R_{Th}}{(\beta + 1) R_E + R_{Th}} \leq 5$$

$$(\beta + 1)(R_E + R_{Th}) \leq 5 [(\beta + 1) R_E + R_{Th}]$$

$$101 R_E + 101 R_{Th} \leq 505 R_E + 5 R_{Th}$$

$$96 R_{Th} \leq 404 R_E$$

$$R_{Th} \leq \frac{404}{96} R_E$$

$$R_{Th} \leq \frac{404}{96} \times 0.96 \text{ k}\Omega$$

$$R_{Th} \leq 4.04 \text{ k}\Omega$$

Let

$$R_{Th} = 4 \text{ k}\Omega$$

$$\begin{aligned} V_{Th} &= R_{Th} I_B + I_E R_E + V_{BE} \\ &= (4 \text{ k}\Omega \times 0.023 \text{ mA}) + 0.7 \text{ V} + (2.32 \text{ mA} \times 0.96 \text{ k}\Omega) \\ &= 3.02 \text{ V} \end{aligned}$$

$$V_{Th} = V_{CC} \frac{R_2}{R_1 + R_2}$$

Multiply and divide by R_1

$\therefore$

$$V_{Th} = \frac{V_{CC}}{R_1} \frac{R_1 R_2}{R_1 + R_2}$$

$$R_{Th} = \frac{R_1 R_2}{R_1 + R_2}$$

$$V_{Th} = \frac{V_{CC}}{R_1} R_{Th}$$

$\therefore$

$$\begin{aligned} R_1 &= \frac{V_{CC}}{V_{Th}} R_{Th} \\ &= \frac{18 \text{ V}}{3.02 \text{ V}} \times 4 \text{ k}\Omega \\ &= 23.84 \text{ k}\Omega \end{aligned}$$

$$R_{Th} = R_1 \parallel R_2$$

$\therefore$

$$\frac{1}{R_{Th}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$\therefore$

$$\frac{1}{R_2} = \frac{1}{R_{Th}} - \frac{1}{R_1} = \frac{R_1 - R_{Th}}{R_{Th} R_1}$$

$$R_2 = \frac{R_1 R_{Th}}{R_1 - R_{Th}}$$

$$\begin{aligned} &= \frac{(23.84 \text{ k}\Omega)(4 \text{ k}\Omega)}{23.84 \text{ k}\Omega - 4 \text{ k}\Omega} \\ &= 4.81 \text{ k}\Omega \end{aligned}$$



Exercise Problems

- 2.1 A *npn* silicon transistor with $V_{BE} = 0.7$ V and $\beta = 75$ is used in a fixed bias circuit. If $R_B = 330$ k Ω , $R_C = 1.5$ k Ω and $V_{CC} = 32$ V. Find the coordinates of the Q point. Draw the dc load line and locate the Q point on dc load line.
- 2.2 A *npn* germanium transistor with $V_{BE} = 0.3$ V and $\beta = 90$ is used in a fixed bias circuit. $V_{CC} = 15$ V, $R_C = 4.7$ k Ω and $R_B = 820$ k Ω . Find the coordinates of the Q point and locate it on the dc load line.
- 2.3 In a fixed bias circuit the coordinates of the Q point which is located at the center of the dc load line $V_{CE} = 5$ V and $I_C = 2$ mA. Find the values of V_{CC} , R_C , and R_B . Assume a silicon transistor with $\beta = 100$.
- 2.4 For a voltage divider-bias circuit, $V_{CC} = 18$ V, $R_1 = 50$ k Ω , $R_2 = 12$ k Ω , $R_C = 2.2$ k Ω , $R_E = 1$ k Ω , $V_{BE} = 0.7$ V and $\beta = 125$. Find the (a) Coordinates of the Q point and locate it on the dc load line (b) Stability factors $S(I_{CO})$ and $S(V_{BE})$.
- 2.5 Find the values of R_1 , R_2 and R_E for a voltage divider bias-circuit given that $V_{CC} = 30$ V, $V_{CE} = 15$ V, $I_C = 3$ mA, $R_C = 3.3$ k Ω , $V_{BE} = 0.7$ V, $\beta = 100$ and $S(I_{CO}) \leq 4$.

Chapter 3

TRANSISTOR AT LOW FREQUENCIES

To analyse the ac operation of a transistor amplifier, it is necessary to develop an ac equivalent circuit for the transistor. This ac equivalent circuit is called the model of the transistor, which simulates the behavior of the transistor, when an ac signal is present. In order to develop the linear model small signal operation is assumed. At low frequencies the junction capacitance of the transistor are not considered due to their high reactance. In this chapter analysis of CE, CB and CC amplifiers with different biasing schemes has been carried out using the r_e model of the transistor. The effects of source resistance and load resistance are also considered.

◆ 3.1 NOTATIONS IN AMPLIFIER ANALYSIS

In an amplifier circuit we come across dc quantities as well as ac quantities. DC voltages and currents result from the biasing network while ac currents and voltages result from the ac input given to the amplifier circuit. To keep dc quantities distinct from ac quantities, the following notations are employed.

DC quantities are represented by upper case letter with uppercase subscripts. For instance:

I_B , I_C and I_E represent dc currents.

V_{BE} , V_{CE} , and V_{CB} represent dc voltages between terminals.

V_B , V_C , and V_E represent dc terminal voltages with respect to ground.

Instantaneous values of ac quantities are represented by lower case letter with lower case subscripts. For instance:

i_b , i_c , and i_e represent instantaneous currents.

v_{be} , v_{ce} , and v_{cb} represent instantaneous voltages between terminals.

v_b , v_c and v_e represent instantaneous terminal voltages with respect to ground.

RMS values of ac quantities are represented by upper case letter with lower case subscripts. For instance:

I_b , I_c and I_e represent rms currents.

V_{be} , V_{ce} and V_{cb} represent rms voltages between terminals.

V_b , V_c and V_e represent rms terminal voltages with respect to ground.

◆ 3.2 AC RESISTANCE OF EMITTER BASE DIODE

In order to use the transistor as an amplifying device, it must be biased properly in the active region. That is the emitter-base junction to be forward biased and collector base junction reverse biased. The forward biased emitter-base junction behaves exactly like a forward biased p - n junction diode.

The ac resistance of the p - n junction diode is given by

$$r_{ac} = \frac{26 \text{ mV}}{I_D} \quad (3.1)$$

where I_D is the dc current through the diode at the Q point. In Equation (3.1), I_D is in milli amperes.

The ac resistance r_e of emitter-base diode can be obtained by replacing I_D by I_E in Equation (3.1).

$$\therefore r_e = \frac{26 \text{ mV}}{I_E} \quad (3.2)$$

I_E is the dc emitter current at the Q point and it is in mill amperes.

◆ 3.3 TWO PORT NETWORK

A two port network as the name implies has two ports: an input port and an output port as shown in Fig. 3.1. A port consists of two terminals.

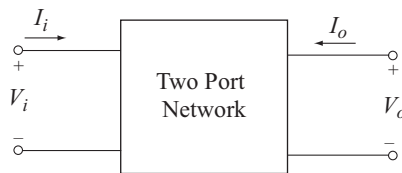


Fig. 3.1 Two port network

A port network is characterized by the two currents, I_i and I_o and the two voltages V_i and V_o .

I_i is the input current

I_o is the output current

V_i is the input voltage

V_o is the output voltage

By convention I_i and I_o are assumed to be into the network.

A BJT has three terminals. It can be modelled as a two port device by making one terminal common to the input and output ports. This leads to the following three configurations of BJT amplifier.

- (a) Common emitter (CE) amplifier
- (b) Common base (CB) amplifier and
- (c) Common collector (CC) amplifier

◆ 3.4 MODELING OF TRANSISTOR

In order to analyze the ac operation of a transistor amplifier, it is necessary to develop an ac equivalent circuit for the transistor. This ac equivalent circuit is called the model of the transistor. The model of the transistor is a combination of circuit elements, properly chosen, the best approximates the actual behavior of the transistor under specific operating conditions.

In order to obtain linear relation between input and output variables, small signal operation is assumed in the development of transistor models. Small signal operation means that, the applied ac input signal causes a small variation in output current and voltages about the Q point.

At low frequencies the junction capacitances of the transistor act as open circuits due to their high reactance. Thus low frequency small signal models do not consider the effect of these junction capacitances.

At high frequencies the junction capacitances conduct appreciably due to their low reactance, providing a feedback path from output to input. Thus high frequency small signal models takes in account the effect of Junction capacitances.

The two most commonly used models of the transistor used in the analysis and design of transistor amplifiers are:

- (a) The hybrid model and
- (b) The r_e model

In the hybrid model, the transistor is modelled based on what is happening at its terminals without regard for the physical process taking place inside the transistor. Transistor data sheets provide the parameters of the hybrid model in their listing, and analysis is simply a matter of inserting the equivalent circuit with the listed values.

The r_e model is the more practical model. The important parameter, r_e of this model is determined by the actual operating conditions rather than using a data sheet value. The r_e model does not include feedback term, which in some cases is important.

It is important to note that the hybrid model and the r_e model are the small signal low frequency models. The hybrid π model (discussed later in the chapter) is used almost exclusively for high frequency analysis. The r_e model is infact a reduced version of the hybrid π model.

◆ 3.5 THE r_e MODEL OF TRANSISTOR IN COMMON BASE (CB) CONFIGURATION

Figure 3.2 shows an npn-transistor in CB configuration represented as a two port network. Since the base terminal is present both in input and output ports, the configuration is common base. It is important to note that, in a transistor the directions of I_b and I_c are same and always opposite to that of I_e . For an npn-transistor I_e is out of the transistor as indicated by the arrow on the emitter terminal and I_b and I_c are into the transistor.

$$\therefore \quad I_i = -I_e \quad (3.3)$$

$$\text{and} \quad I_o = I_c \quad (3.4)$$